

Chapter 1

Revision

$$V = IR$$

$$I = Q/t$$

Electric Current & Ohm's Law

100

QUESTIONS

67

FIGURES

4

LESSONS

100%

STEP-BY-STEP

Lesson 1

Electric current & Ohm's law

Lesson 2

Connecting resistors

Lesson 3

Ohm's law for a closed circuit

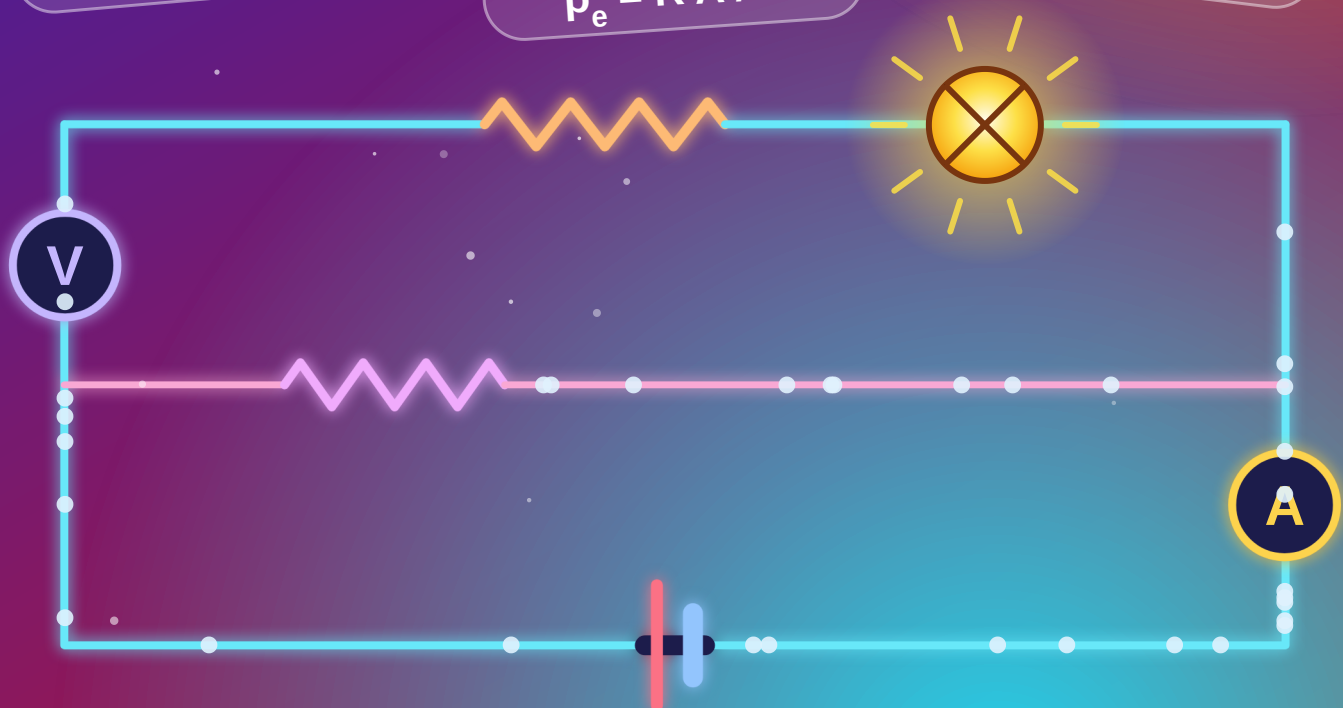
Lesson 4

Kirchhoff's laws

$$V_B = I(R + r)$$

$$\rho_e = RA/L$$

$$\Sigma I = 0$$



Mr. Gemy

PHYSICS

Final answers + full step-by-step solutions at the end of the booklet

Data & notes for all questions

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$\pi = 3.14 \text{ (unless } 22/7 \text{ is given)}$$

ammeters : negligible resistance

voltmeters : very large resistance

$$r = 0 \rightarrow \text{negligible internal resistance}$$

MCQ

Problem — write the steps

★ HOTS — higher order thinking

Section 1 — Electric Current & Ohm's Law

Questions 1 – 25

Lesson 1 · current · p.d. · resistance · resistivity · energy & power

- 1 2×10^{20} electrons pass through a cross-section of a wire in 8 s. The intensity of the electric current in the wire equals

MCQ

A 4 A

B 0.25 A

C 2.5 A

D 40 A

- 2 The graph shows the quantity of charge (Q) passing through a cross-section of a conductor against time (t). The ratio between the current in part ab and the current in part bc (I_{ab} / I_{bc}) is

MCQ

A 3

B $1/3$

C 2

D 6

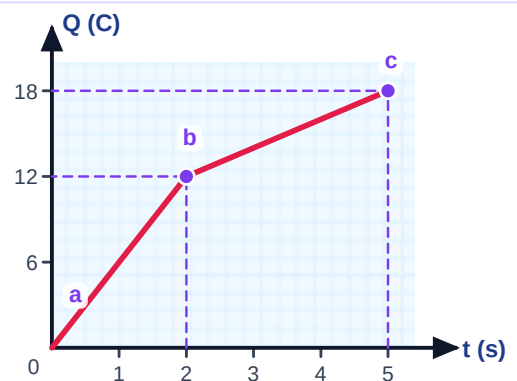


Fig. 1

- 3 The graph shows the current (I) in a conductor against time (t). The number of electrons that pass through a cross-section of the conductor during the first 10 s is

MCQ

A 3.125×10^{19}

B 1.25×10^{20}

C 1.875×10^{20}

D 4.8×10^{-18}

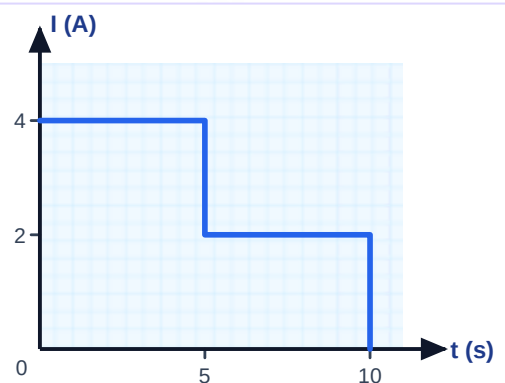


Fig. 2

4 The work done to transfer 5×10^{19} electrons between two points of a conductor is 48 J. The potential difference between the two points equals

MCQ

A 8 V

B 6 V

C 0.6 V

D 384 V

5 The potential difference across the filament of a lamp is 12 V and the resistance of the filament is 4Ω . **Calculate** the number of electrons that pass through a cross-section of the filament in one minute.

Problem

6 The graph shows the relation between the potential difference (V) and the current (I) for two conductors x and y. The ratio of their resistances (R_x / R_y) is

★ HOTS

A $1/3$

B $\sqrt{3}$

C 3

D $1/\sqrt{3}$

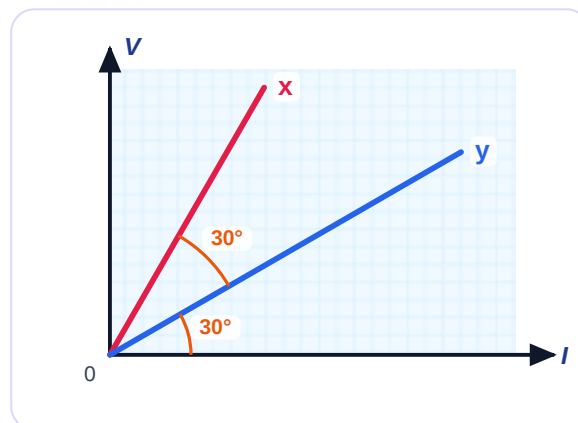
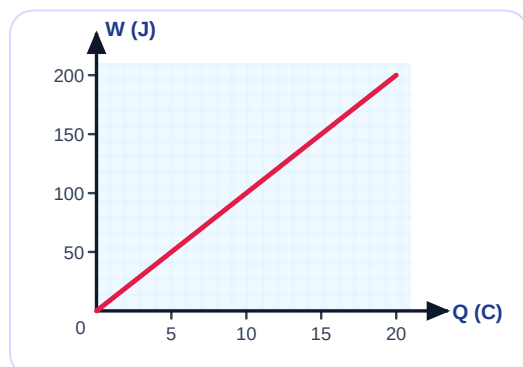


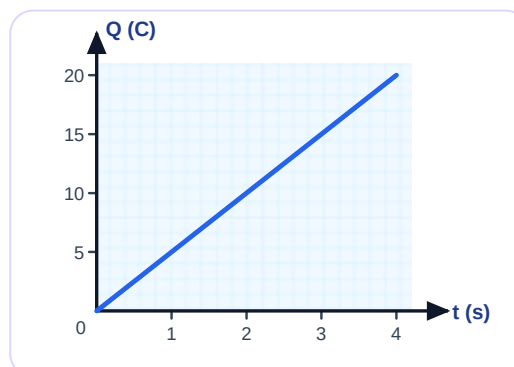
Fig. 3

7 Graph (1) shows the work done (W) to move charge (Q) through a conductor, and graph (2) shows the charge passing through the same conductor against time. The resistance of the conductor equals

MCQ



Graph (1)



Graph (2)

Fig. 4

A 0.5Ω

B 2Ω

C 5Ω

D 15Ω

8 A copper wire has a length of 20 m and a cross-sectional area of 0.5 mm^2 . If the resistivity of copper is $1.7 \times 10^{-8} \Omega \cdot \text{m}$, the resistance of the wire equals

MCQ

A 0.68 Ω

B 6.8 Ω

C 0.068 Ω

D 0.34 Ω

9 A wire of length 6.28 m and radius 1 mm is made of a material of resistivity $5 \times 10^{-7} \Omega \cdot \text{m}$. It is connected across a 3 V battery of negligible internal resistance. The current in the wire is

MCQ

A 1.5 A

B 0.33 A

C 6 A

D 3 A

10 A metal wire is stretched uniformly until its length is doubled (its volume stays constant). Which row shows what happens to its resistance and to the conductivity of its material?

MCQ

A R doubles , σ unchanged

B R quadruples , σ unchanged

C R quadruples , σ halves

D R unchanged , σ doubles

11 A wire of resistance 5Ω is drawn through a die so that its radius becomes half its original value (the volume stays constant). Its new resistance is

★ HOTS

A 10 Ω

B 20 Ω

C 40 Ω

D 80 Ω

12 Two wires x and y are made of the same metal. Wire x has a length of $3L$ and a mass of 4 g, while wire y has a length L and a mass of 12 g. The ratio of their resistances (R_x / R_y) equals

★ HOTS

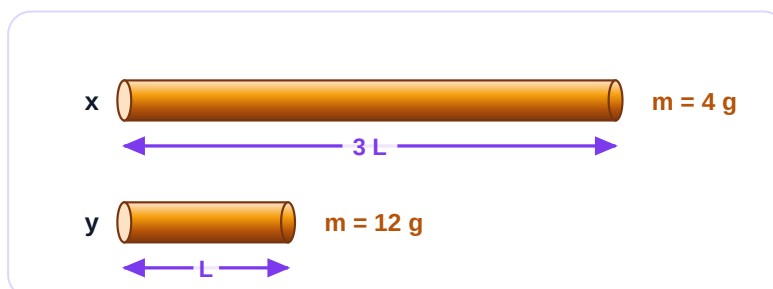


Fig. 5

A 9

B 27

C 3

D $1/27$

Problem

- 13 A wire of cross-sectional area 0.2 mm^2 and resistivity $5 \times 10^{-7} \Omega \cdot \text{m}$ is wound into a circular coil of 100 turns, each of radius 7 cm. **Find:** (i) the length of the wire, (ii) its resistance, (iii) the current in the coil when it is connected to 220 V. ($\pi = 22/7$)

Problem

- 14 The graph shows how the resistance (R) of a uniform wire changes with its length (L). The cross-sectional area of the wire is 0.2 mm^2 . **Find:** (i) the resistivity of its material, (ii) the length of this wire that has a resistance of 4.5Ω .

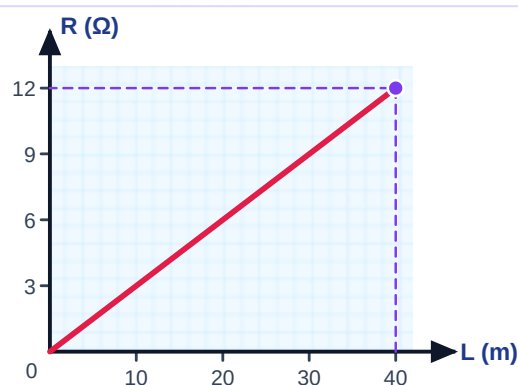


Fig. 6

MCQ

- 15 The temperature of a metal wire connected across a constant potential difference is raised. Which row is correct?

A R increases , I decreases , σ decreases

B R increases , I increases , σ decreases

C R decreases , I increases , σ increases

D R increases , I decreases , σ unchanged

MCQ

- 16 The figure shows four wires a, b, c and d made of the same metal (their lengths and cross-sectional areas are marked). When each is connected alone across the same potential difference, the correct order of the currents is

A $a > b > c > d$

B $c > d > a > b$

C $d > c > a > b$

D $b > a > d > c$

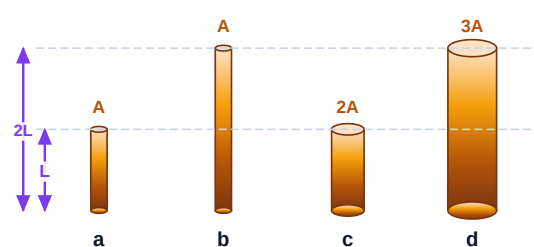


Fig. 7

- 17 A hollow copper tube has a length of 2 m, an inner radius of 2 mm and an outer radius of 3 mm. The resistivity of its material is $3.14 \times 10^{-8} \Omega \cdot \text{m}$. The resistance between its two ends is

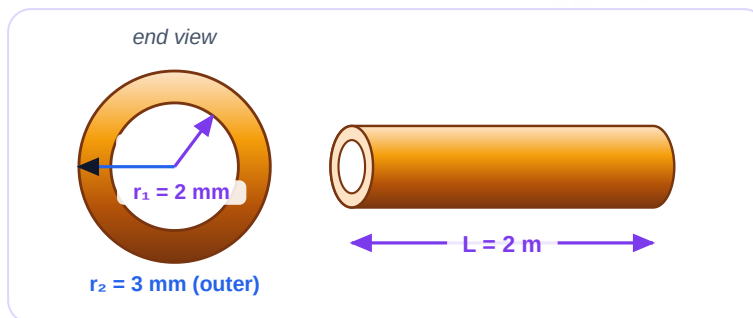


Fig. 8

- A $2 \times 10^{-3} \Omega$ B $4 \times 10^{-3} \Omega$
 C $8 \times 10^{-3} \Omega$ D $1.6 \times 10^{-3} \Omega$

- 18 A conductor of length 1.57 m has a semicircular cross-section of diameter 4 mm, as shown. If the resistivity of its material is $2 \times 10^{-6} \Omega \cdot \text{m}$, its resistance equals

MCQ

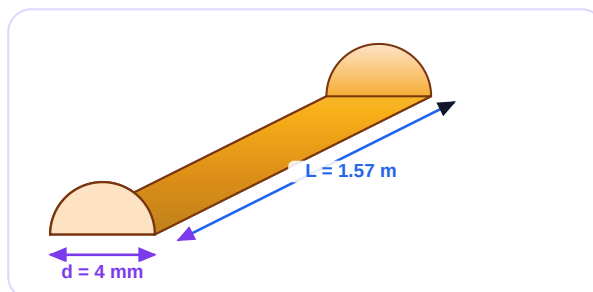


Fig. 9

- A 0.25 Ω B 1 Ω
 C 0.5 Ω D 2 Ω

- 19 A lamp is labelled (220 V , 100 W). If it is connected to a 110 V source (its resistance stays constant), the power it consumes is

MCQ

- A 50 W B 25 W
 C 100 W D 12.5 W

- 20 The graph shows the power (P) consumed in a resistor against the square of the current (I^2). If this resistor is connected alone across a 10 V source, the power it consumes is

MCQ

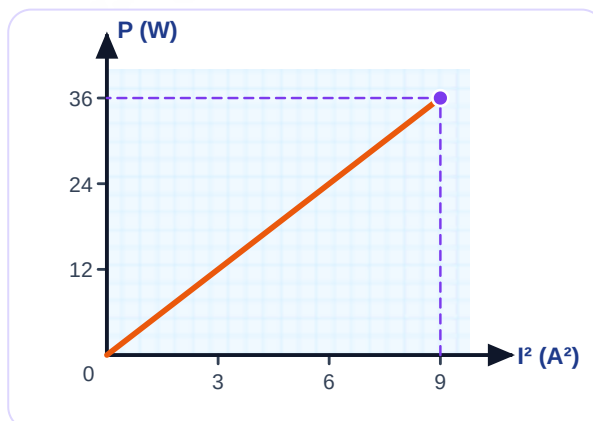


Fig. 10

- A 25 W B 40 W
 C 2.5 W D 400 W

21 An electric heater of resistance $22\ \Omega$ works on 220 V for 10 minutes. The electric energy it consumes is

MCQ

A $2.2 \times 10^3\text{ J}$

B $1.32 \times 10^5\text{ J}$

C $1.32 \times 10^6\text{ J}$

D $2.64 \times 10^6\text{ J}$

22 Two wires of the same metal and the same length have radii in the ratio $1 : 2$. They are connected in parallel across the same source. The ratio of the powers they consume (P_1 / P_2) is

MCQ

A $1/2$

B 2

C $1/4$

D 4

23 In the circuit shown, the battery has negligible internal resistance. Which graph shows the **ammeter** reading against the resistance R_v taken from the rheostat?

MCQ

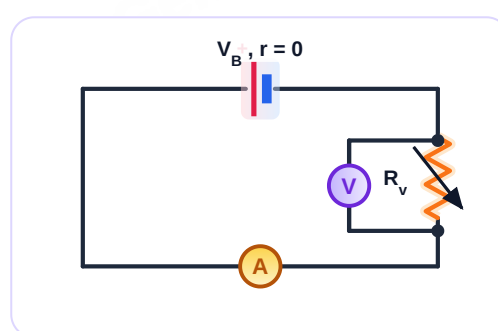
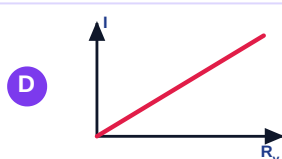
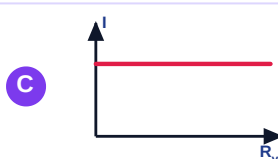
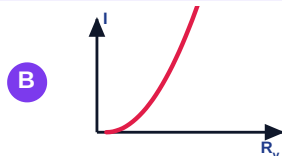
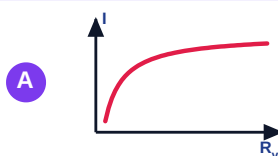
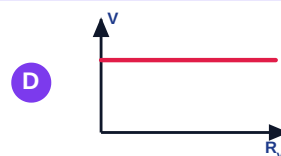
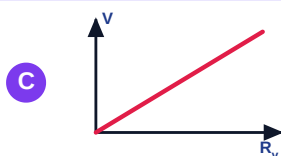
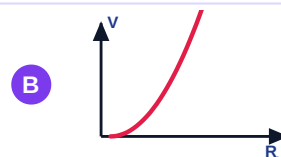
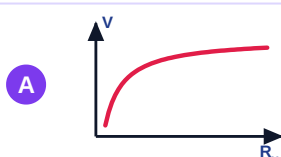


Fig. 11



24 For the circuit of Fig. 11, which graph shows the **voltmeter** reading against R_v ?

MCQ



25 Conductor x has a length L , a cross-sectional area A and a conductivity σ . Conductor y has a length $2L$, a cross-sectional area $3A$ and a conductivity $\sigma/2$. When each is connected across the same potential difference, the ratio of the currents (I_x / I_y) is

★ HOTS

A $4/3$

B $3/4$

C 3

D $1/3$

Section 2 — Connecting Resistors

Lesson 2 · series · parallel · networks · lamps & switches

Questions 26 – 51

- 26 Four identical resistors are connected in the four ways shown. The correct order of the equivalent resistances from the **largest** to the **smallest** is

MCQ

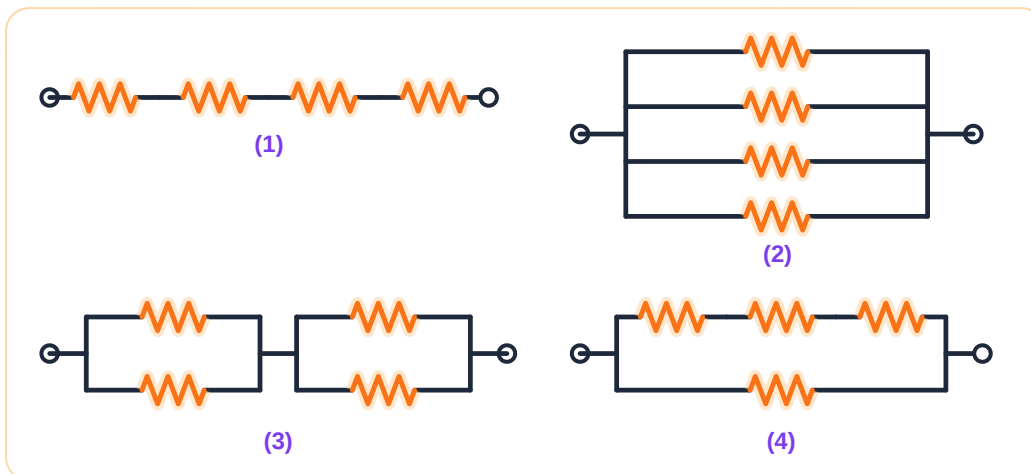


Fig. 12

A (1), (3), (4), (2)

B (2), (4), (3), (1)

C (1), (4), (3), (2)

D (3), (1), (2), (4)

- 27 A uniform wire of resistance $90\ \Omega$ is cut into six equal parts, which are connected as shown. The equivalent resistance between a and b is

MCQ

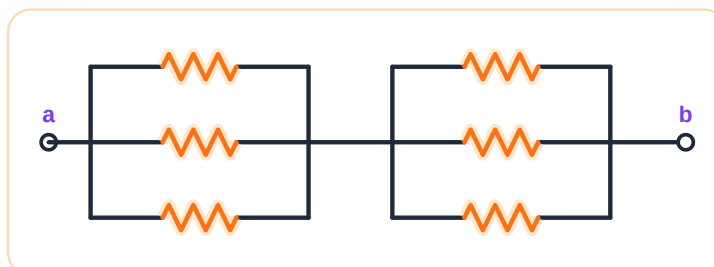


Fig. 13

A $15\ \Omega$

B $5\ \Omega$

C $10\ \Omega$

D $30\ \Omega$

- 28 A current of $2\ \text{A}$ enters two resistors of $6\ \Omega$ and $3\ \Omega$ connected in parallel. The current I_1 in the $6\ \Omega$ resistor is

MCQ

A $4/3\ \text{A}$

B $2/3\ \text{A}$

C $1\ \text{A}$

D $2\ \text{A}$

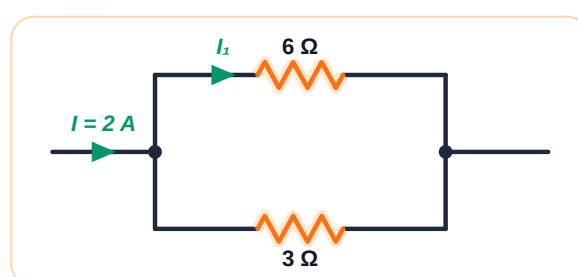


Fig. 14

- 29 In the network shown, the ends of the $4\ \Omega$ resistor are joined by a connecting wire of negligible resistance. The equivalent resistance between a and b is

MCQ

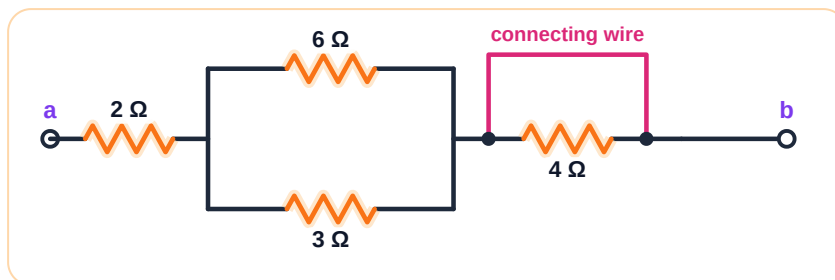


Fig. 15

A $8\ \Omega$

B $4\ \Omega$

C $6\ \Omega$

D $2\ \Omega$

- 30 In the network shown, the equivalent resistance between a and b is

★ HOTS

A $4.5\ \Omega$

B $2.5\ \Omega$

C $15\ \Omega$

D $3.6\ \Omega$

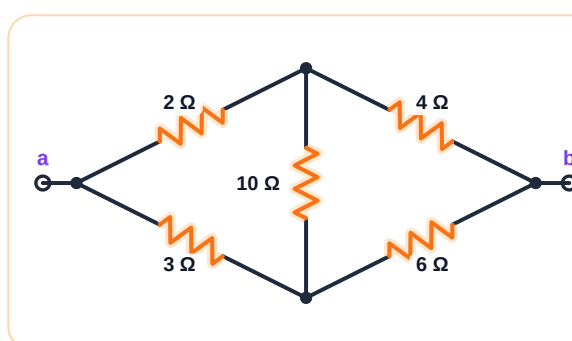


Fig. 16

- 31 In the circuit shown, the reading of the voltmeter is

MCQ

A $4\ \text{V}$

B $6\ \text{V}$

C $8\ \text{V}$

D $12\ \text{V}$

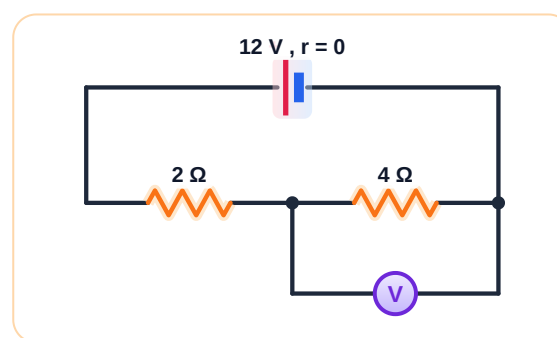


Fig. 17

- 32 In the circuit shown, the voltmeter reads $4\ \text{V}$. The current in the $6\ \Omega$ resistor is

MCQ

A $0.8\ \text{A}$

B $1.2\ \text{A}$

C $1\ \text{A}$

D $2\ \text{A}$

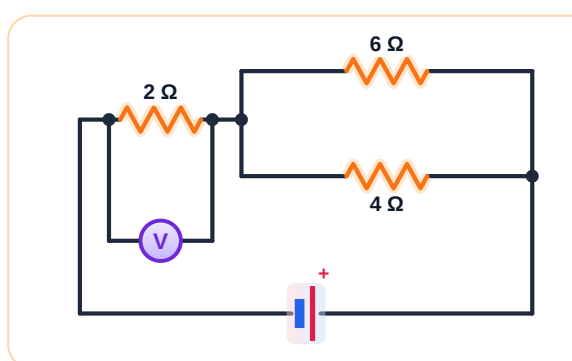


Fig. 18

33 In the circuit shown, the four lamps are identical. When switch K is closed, what happens to the brightness of lamps A and B?

★ HOTS

A A increases , B decreases

B A decreases , B increases

C both increase

D A unchanged , B decreases

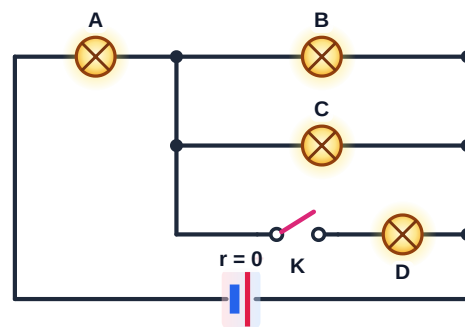


Fig. 19

34 Two lamps rated (220 V , 100 W) and (220 V , 60 W) are connected **in series** to a 220 V source. The brighter lamp is

MCQ

A the 100 W lamp

B the 60 W lamp

C both are equally bright

D neither lamp lights

35 Two identical resistors connected in series to a source of constant p.d. consume a total power P. If they are connected in parallel to the same source, the total power becomes

MCQ

A $P / 4$

B $2P$

C $4P$

D P

36 In the network shown, the equivalent resistance between a and b is

MCQ

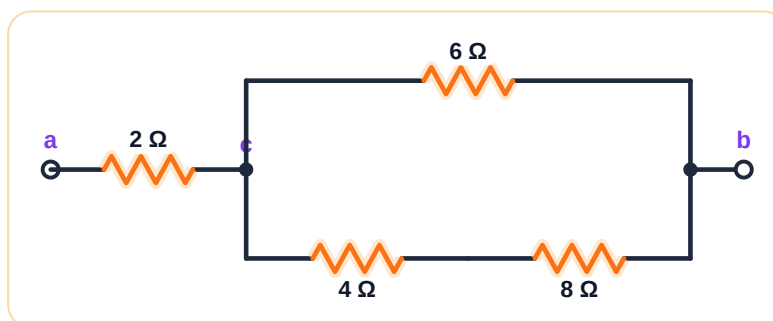


Fig. 20

A 6Ω

B 20Ω

C 8Ω

D 4Ω

- 37 Conductor x (length L , area A , conductivity 4σ) lies in perfect contact along the first half of conductor y (length $2L$, area $2A$, conductivity σ), as shown. If the resistance of x is $5\ \Omega$, the resistance between a and b is

★ HOTS

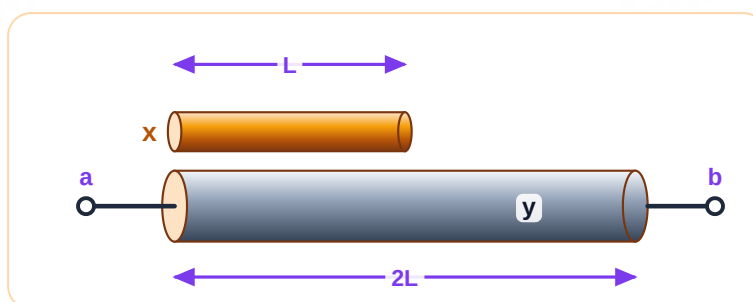


Fig. 21

- A $7.5\ \Omega$ B $15\ \Omega$
C $20\ \Omega$ D $13.3\ \Omega$

- 38 Using **all three** of three identical $6\ \Omega$ resistors, which of the following values **cannot** be obtained?

MCQ

- A $18\ \Omega$ B $9\ \Omega$
C $4\ \Omega$ D $12\ \Omega$

- 39 In the circuit shown, the ratio of the voltmeter readings (V_1 / V_2) is

MCQ

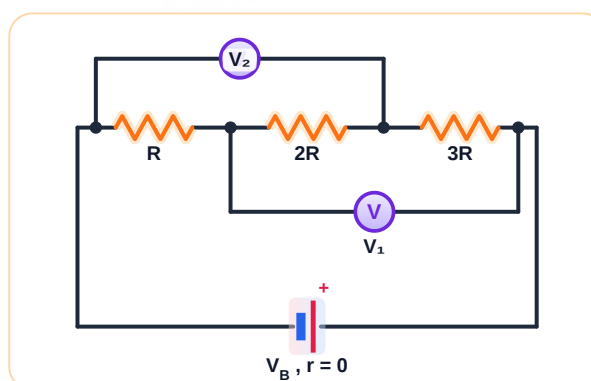


Fig. 22

- 40 In the circuit shown, **find**: (i) the total current, (ii) the current in each of the $6\ \Omega$ and $12\ \Omega$ resistors, (iii) the power consumed in the $4\ \Omega$ resistor.

Problem

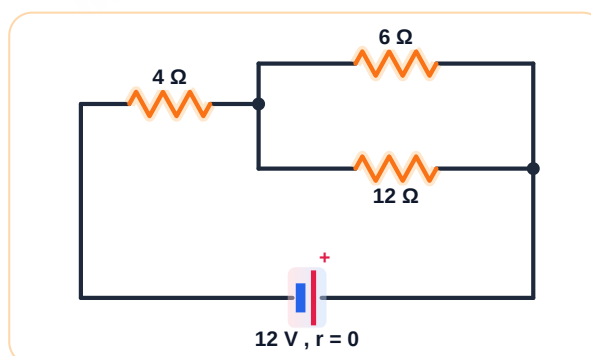


Fig. 23

- 41 In the $V - I$ graph (the marked point is on line 4), lines 1 and 3 belong to two resistors x and y . Which line can represent x and y connected **in parallel**?

- A line 1
B line 2
C line 3
D line 4

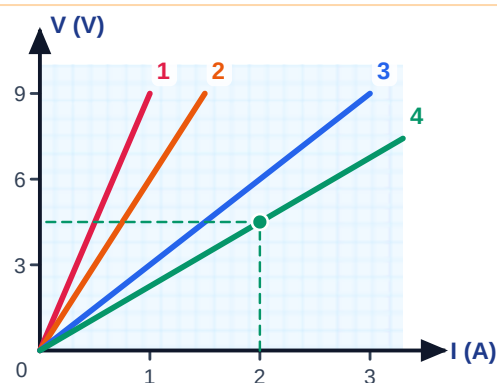


Fig. 24

- 42 In the circuit shown, when switch K is closed, the ammeter reading changes

- A from 1 A to 3 A
B from 3 A to 1 A
C from 1 A to 1.5 A
D it stays 1 A

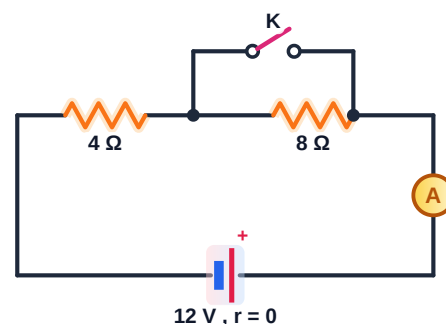


Fig. 25

- 43 In the circuit shown, the ammeter reads 4 A. **Find** the value of the resistance R .

Problem

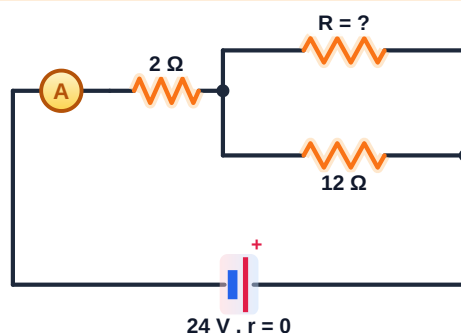


Fig. 26

- 44 In the circuit shown, the three lamps are identical. If lamp C burns out (its filament breaks), what happens to the brightness of lamps A and B ?

- A A increases , B decreases
B both decrease
C A decreases , B increases
D both go off

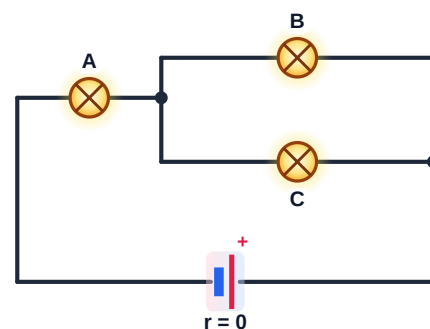


Fig. 27

- 45 A uniform wire of resistance $16\ \Omega$ is bent into a circle. The resistance between points a and b, which are a quarter of the circle apart, is

A $3\ \Omega$

B $4\ \Omega$

C $8\ \Omega$

D $12\ \Omega$

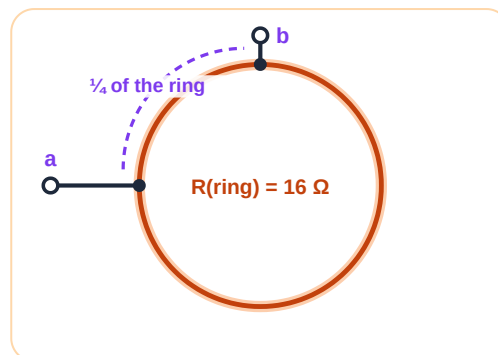


Fig. 28

- 46 In the network shown, the equivalent resistance between a and b is

A $15\ \Omega$

B $1.5\ \Omega$

C $2\ \Omega$

D $3\ \Omega$

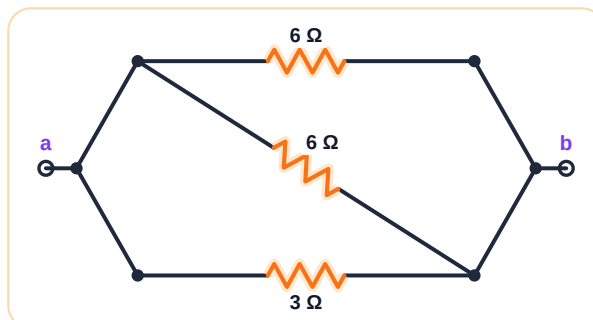


Fig. 29

- 47 n identical resistors give an equivalent resistance of $50\ \Omega$ when connected in series and $2\ \Omega$ when connected in parallel. **Find** n and the resistance of each resistor.

Problem

- 48 In the circuit shown, when switch K is closed, which row is correct?

MCQ

A ammeter reading increases, $I(6\ \Omega)$ unchanged

B ammeter reading unchanged, $I(6\ \Omega)$ decreases

C ammeter reading increases, $I(6\ \Omega)$ increases

D ammeter reading decreases, $I(6\ \Omega)$ unchanged

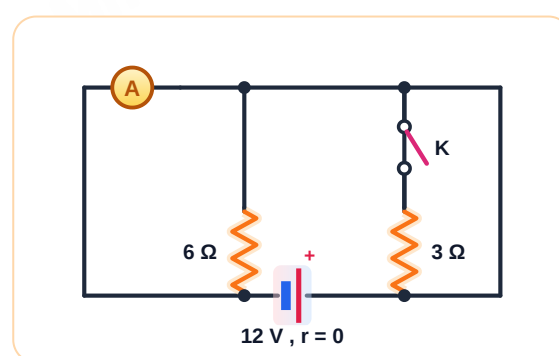


Fig. 30

- 49 Three resistors of $2\ \Omega$, $3\ \Omega$ and $6\ \Omega$ are connected in parallel to a $6\ \text{V}$ battery of negligible internal resistance. **Find** the power consumed in each resistor and the total power.

Problem

50 In the circuit shown, the resistance taken from the rheostat is **decreased**. Which row is correct?

MCQ

A lamp brighter ,
voltmeter reading
increases

B lamp dimmer ,
voltmeter reading
increases

C lamp brighter ,
voltmeter reading
decreases

D lamp unchanged ,
voltmeter reading
unchanged

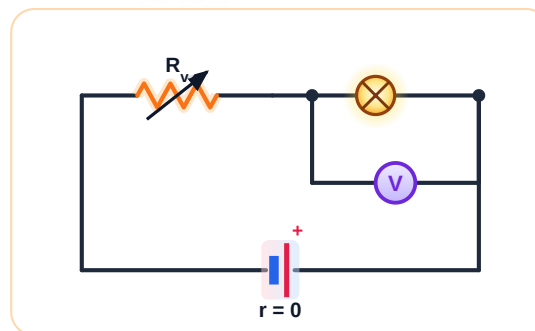


Fig. 31

51 A uniform wire of resistance $36\ \Omega$ is bent into an equilateral triangle xyz. The resistance between two corners x and y is

MCQ

A $8\ \Omega$

B $12\ \Omega$

C $18\ \Omega$

D $24\ \Omega$

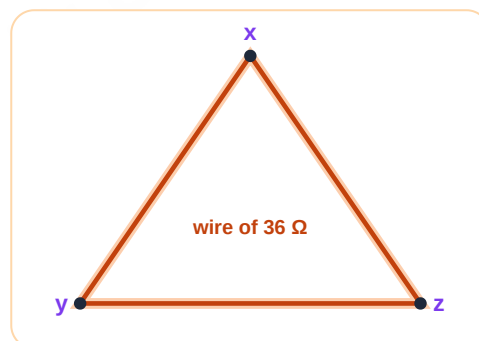


Fig. 32

Section 3 — Ohm's Law for the Closed Circuit

Lesson 3 · emf · internal resistance · terminal p.d. · batteries together

Questions 52 – 75

52 A battery of emf 12 V and internal resistance r is being **charged** by a current I . The potential difference between its terminals is

MCQ

A less than 12 V

B equal to 12 V

C more than 12 V

D zero

53 In the circuit shown, the reading of the voltmeter is

MCQ

A 12 V

B 10 V

C 2 V

D 8 V

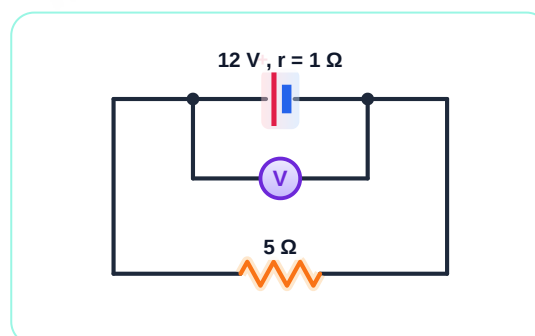


Fig. 33

- 54** In the circuit shown, the voltmeter reads 12 V when switch K is open and 10 V when K is closed, while the ammeter then reads 2 A. **Find:** (i) the emf of the battery, (ii) its internal resistance, (iii) the resistance R.

Problem

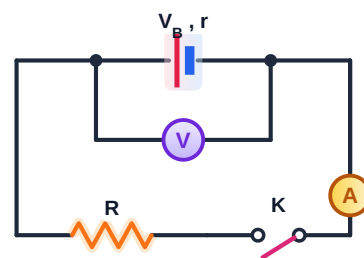


Fig. 34

- 55** A battery gives a current of 2 A when it is connected to a $5\ \Omega$ resistor, and a current of 1 A when it is connected to an $11\ \Omega$ resistor. **Find** its emf and its internal resistance.

Problem

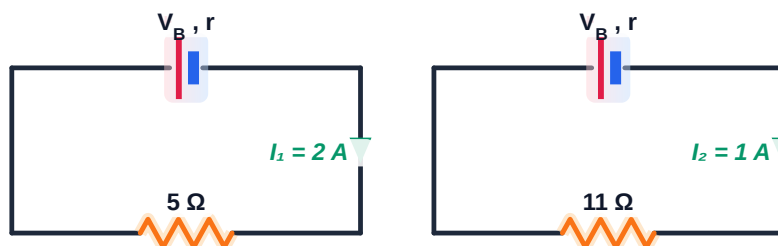


Fig. 35

- 56** The voltmeter across the terminals of a battery reads V_0 when no current flows, and $0.75 V_0$ when an external resistor R is connected. The ratio R / r is

MCQ

A $1/3$

B 3

C 4

D 0.75

- 57** Two batteries (6 V , $0.5\ \Omega$) and (9 V , $1\ \Omega$) are connected in series, supporting each other, with a $3.5\ \Omega$ resistor. **Find:** (i) the current, (ii) the p.d. across each battery.

Problem

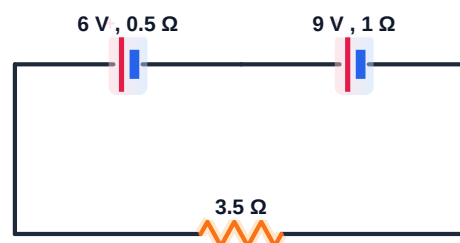


Fig. 36

- 58 A battery (12 V , $1\ \Omega$) and a battery (6 V , $1\ \Omega$) are connected in **opposition** with a $4\ \Omega$ resistor. The potential difference across the terminals of the 6 V battery is

A 5 V B 6 V
C 7 V D 11 V

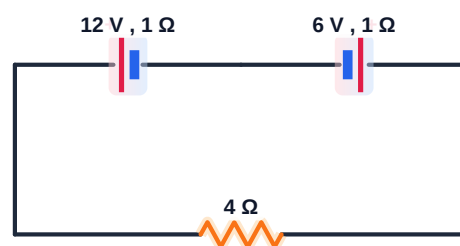


Fig. 37

- 59 The graph shows the terminal potential difference (V) of a battery against the current (I) it delivers. The internal resistance of the battery is

A $2\ \Omega$ B $0.5\ \Omega$
C $6\ \Omega$ D $12\ \Omega$

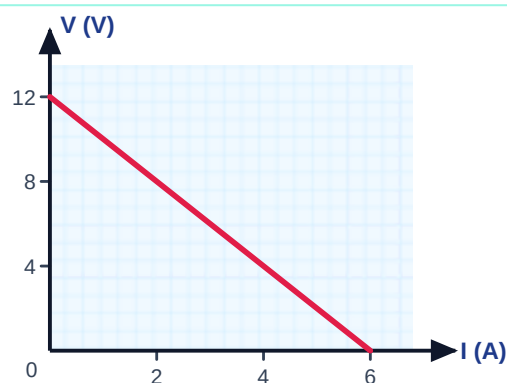


Fig. 38

- 60 For the battery of Fig. 38, the current it delivers to an external resistor of $4\ \Omega$ is

A 3 A B 1.5 A
C 2 A D 6 A

- 61 In the circuit shown, when the resistance taken from the rheostat is **increased**, the reading of the voltmeter across the battery

A increases B decreases
C stays constant D becomes zero

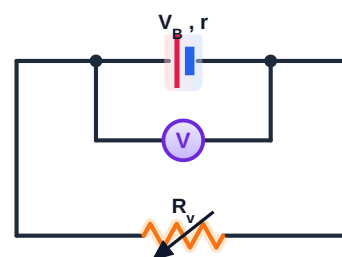


Fig. 39

- 62 Four identical cells, each of emf 1.5 V and internal resistance $0.5\ \Omega$, are connected in series with a $4\ \Omega$ resistor. The current in the circuit is

A 1.5 A B 0.75 A
C 1 A D 3 A

★ HOTS

- 63 Six identical cells (2 V , $0.5\ \Omega$ each) are connected in series with a $5\ \Omega$ resistor, but one cell is connected the wrong way round, as shown. The current in the circuit is

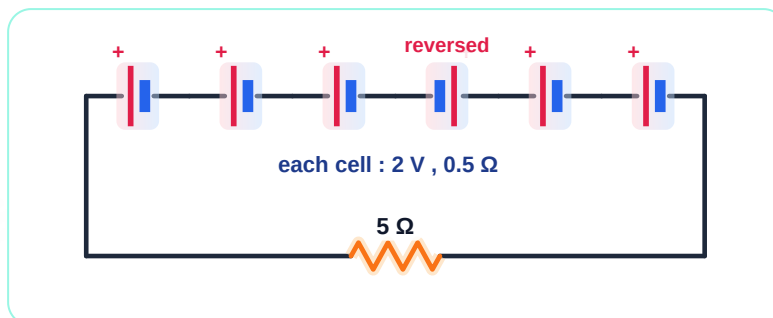


Fig. 40

A 1.5 A B 1 A C 0.5 A D 2 A

- 64 In the circuit shown, the five resistors are identical and the battery has an internal resistance of $1\ \Omega$. The voltmeter reads 12 V when K is open and 9 V when K is closed. The value of R is

★ HOTS

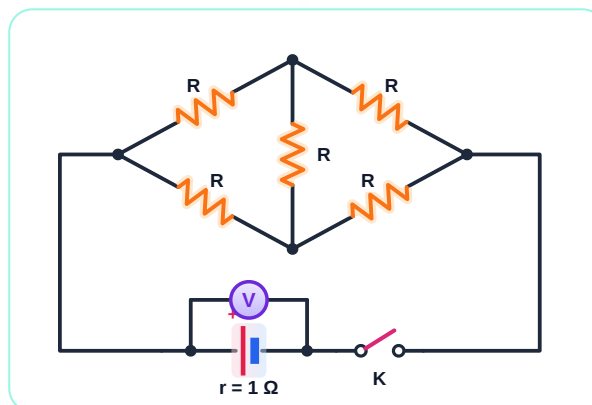
A $1.5\ \Omega$ B $6\ \Omega$ C $4\ \Omega$ D $3\ \Omega$ 

Fig. 41

- 65 A battery of emf V_B and internal resistance r is short-circuited by a thick wire of negligible resistance. The voltmeter connected across its terminals reads

MCQ

A V_B B $V_B / 2$

C zero

D more than V_B

★ HOTS

- 66 In the circuit shown, the ammeter A_2 reads zero. The reading of the ammeter A_1 is

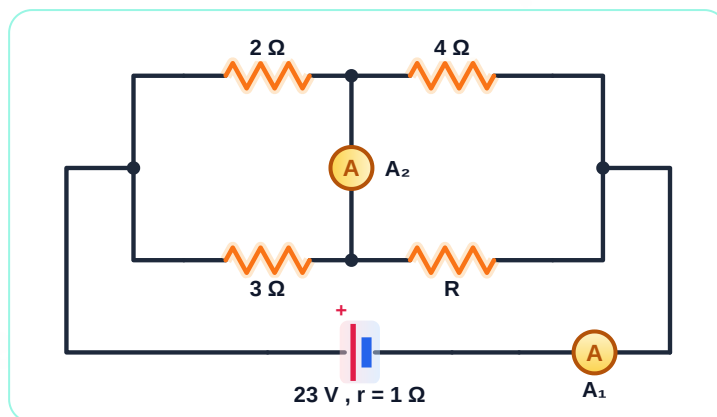


Fig. 42

A 4 A

B 5 A

C 6.4 A

D 2.5 A

- 67 In the circuit shown, the $6\ \Omega$ resistor carries 1 A. Is the internal resistance of the 15 V battery considered or neglected? **Explain** with calculation.

Problem

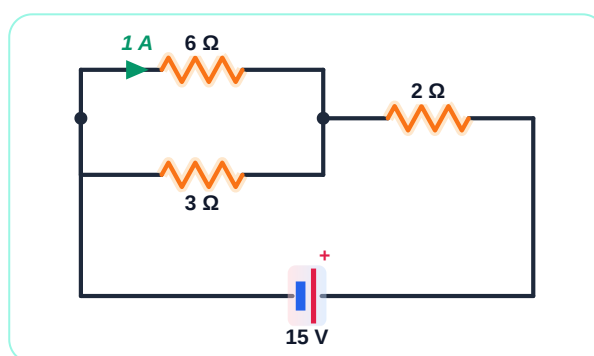


Fig. 43

- 68 Figures (1) and (2) are parts of two closed circuits; in each, a current of 2 A flows in the direction shown. The ratio V_{xy} / V_{zk} is

★ HOTS

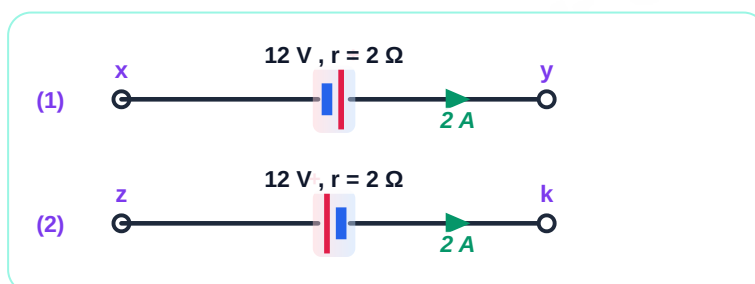


Fig. 44

A $1/2$

B 2

C 1

D $3/4$

69 A battery of emf 24 V and internal resistance $1\ \Omega$ is connected to a $7\ \Omega$ resistor. The heat produced inside the battery in one minute is

MCQ

A 540 J

B 9 J

C 3780 J

D 4320 J

70 In the circuit shown, $R_1 = 3r$ and the voltmeter V across the battery reads four times the voltmeter V_1 across R_1 . Find R_2 in terms of r .

Problem

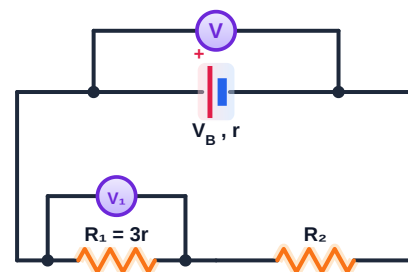


Fig. 45

71 A battery is connected to a variable resistor R and the graph of $1/I$ against R is plotted. The emf and the internal resistance of the battery are

★ HOTS

A 4 V , $2\ \Omega$

B 2 V , $4\ \Omega$

C 4 V , $0.5\ \Omega$

D 0.25 V , $2\ \Omega$

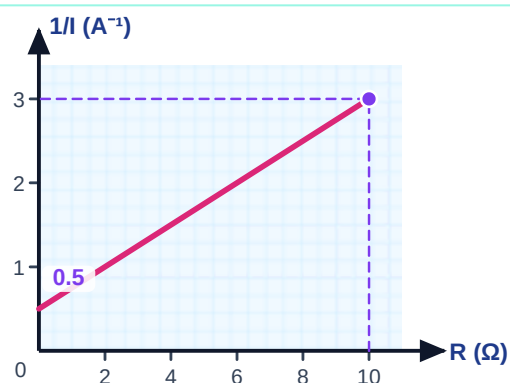


Fig. 46

72 A battery of emf 6 V and internal resistance $0.5\ \Omega$ is charged by a current of 2 A. The potential difference across its terminals is

MCQ

A 5 V

B 6 V

C 7 V

D 8 V

73 The largest current that a battery of emf 9 V and internal resistance $0.5\ \Omega$ can deliver (when it is short-circuited) is

MCQ

A 4.5 A

B 9 A

C 18 A

D infinite

- 74** A lamp labelled (6 V , 3 W) is connected in series with a rheostat to a battery of emf 9 V and internal resistance 1 Ω . **Find** the resistance that must be taken from the rheostat for the lamp to work normally.

Problem

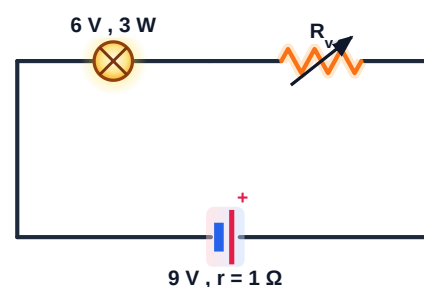


Fig. 47

- 75** A battery delivers the same power to an external resistor of 4 Ω as to one of 9 Ω . **Find** its internal resistance.

★ HOTS

Section 4 — Kirchhoff's Laws

Lesson 4 · junction rule · loop rule · multi-loop circuits · potentials

Questions 76 – 100

- 76** The figure shows five wires meeting at junction a. The current I in wire c is

MCQ

- A** 3 A , from a to c
B 5 A , from c to a
C 5 A , from a to c
D 23 A , from c to a

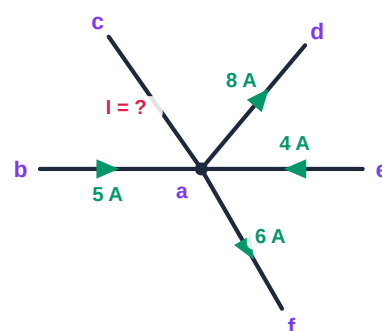


Fig. 48

- 77** Kirchhoff's first law and Kirchhoff's second law are based on the conservation of

MCQ

- A** energy , charge
B charge , energy
C charge , mass
D mass , energy

78 In the figure, the current in the $6\ \Omega$ resistor is 2 A. The current I is

MCQ

A 1 A

B 2 A

C 4 A

D 3 A

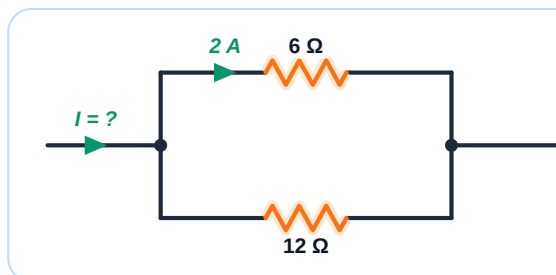


Fig. 49

79 When a closed loop is followed through a resistor R in the same direction as the current I , the change in potential is written in the loop equation as

MCQ

A $+IR$

B $-IR$

C zero

D $+V_B$

80 In the circuit shown, both batteries have negligible internal resistance. Use Kirchhoff's laws to find the currents I_1 , I_2 and I_3 .

★ HOTS

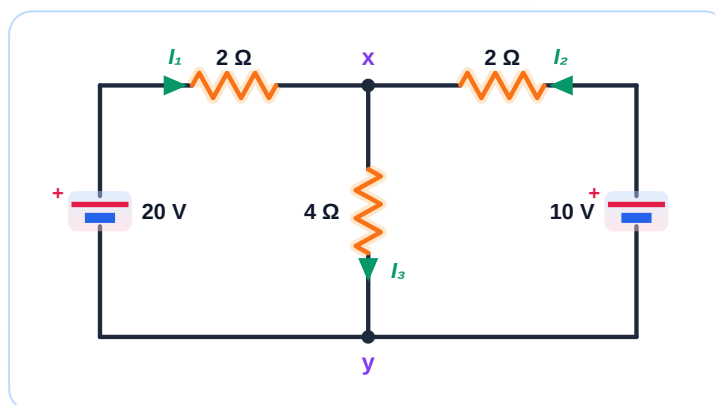


Fig. 50

81 For the circuit of Fig. 50, the power consumed in the $4\ \Omega$ resistor is

MCQ

A 36 W

B 12 W

C 64 W

D 9 W

82 For the circuit of Fig. 50, which statement about the 10 V battery is correct?

MCQ

A it discharges with a current of 1 A

B it is being charged with a current of 1 A

C it carries no current

D it discharges with a current of 3 A

83 In the circuit shown, the current in the $1\ \Omega$ resistor is

MCQ

A 4.5 A

B 1.5 A

C 6 A

D 3 A

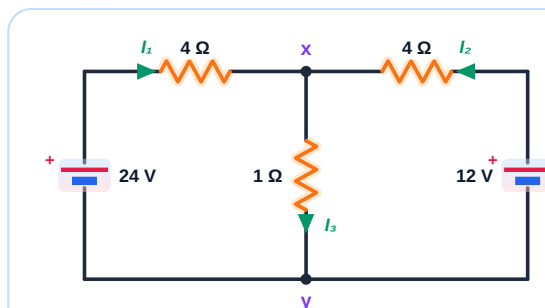


Fig. 51

84 Between points a and b, a $2\ \Omega$ resistor carries 3 A in the direction shown, followed by a 10 V battery ($r = 0$). The potential difference $V_{ab} = V_a - V_b$ is

★ HOTS



Fig. 52

A 16 V

B 4 V

C -4 V

D -16 V

85 Twelve identical resistors, each R , form the edges of a cube. The equivalent resistance between two opposite corners a and b is

★ HOTS

A $R/2$

B $5R/6$

C $3R/2$

D R

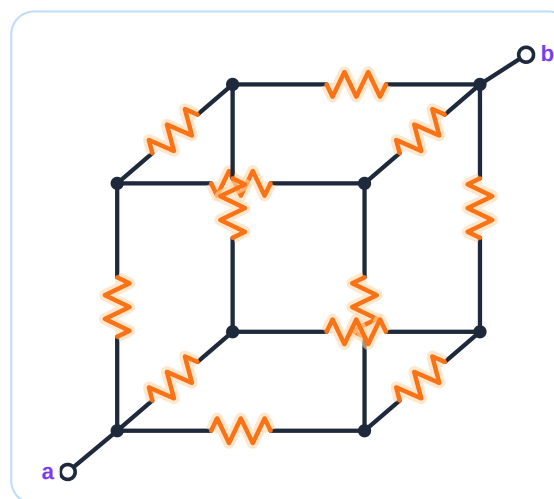


Fig. 53

- 86 In the network shown, the battery has an emf of 14 V and negligible internal resistance. Use Kirchhoff's laws to **find**: (i) the current in the $1\ \Omega$ resistor CD, (ii) the current from the battery, (iii) the equivalent resistance of the network.

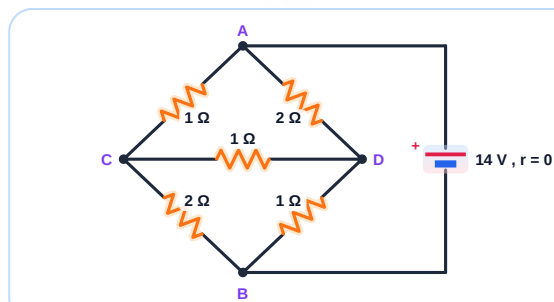


Fig. 54

- 87 In the circuit shown, the $2\ \Omega$ resistor carries 6 A. **Find** I_1 , I_2 and R_2 .

Problem

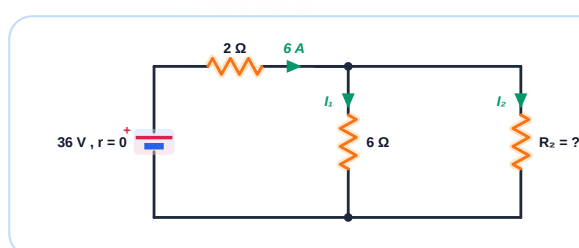


Fig. 55

- 88 In the circuit shown, the galvanometer G reads zero. The emf V_{B2} of the second battery equals

★ HOTS

- A 4 V B 12 V
C 8 V D 6 V

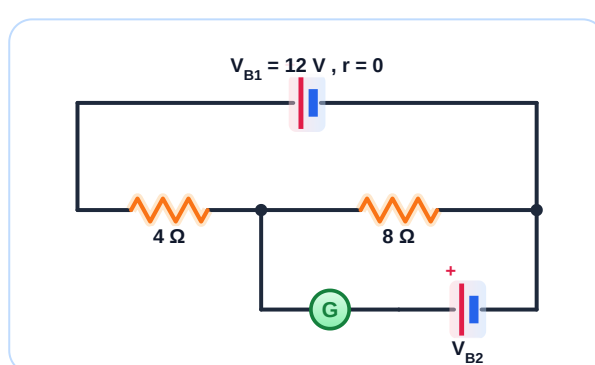


Fig. 56

- 89 Two identical batteries (6 V, $1\ \Omega$) are connected in parallel, as shown, to a $2.5\ \Omega$ resistor. The current in each battery is

MCQ

- A 2 A B 1 A
C 2.4 A D 0.5 A

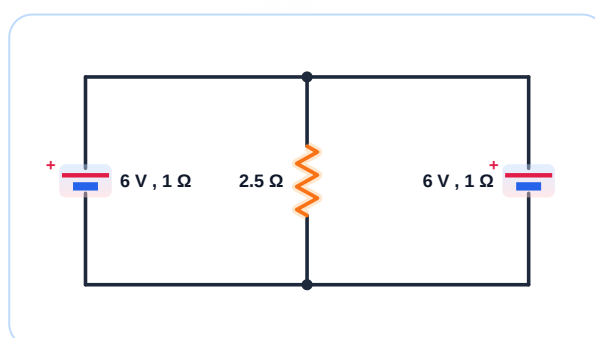


Fig. 57

- 90 In the circuit shown, the three batteries are identical ($r = 0$) and all resistors equal R . When both switches K_1 and K_2 are closed, the power consumed in S becomes n times its value when both are open, where n equals

★ HOTS

A $3/2$

B $9/4$

C 3

D $9/16$

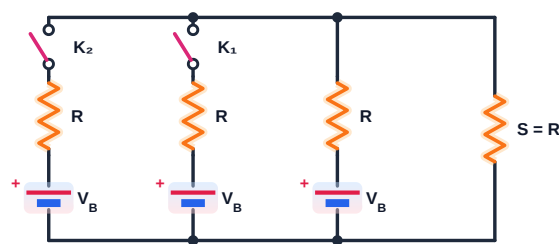


Fig. 58

- 91 In the single loop shown (all batteries have $r = 0$), the current in the circuit is

MCQ

A 4.4 A

B 2 A

C 1.2 A

D 3.2 A

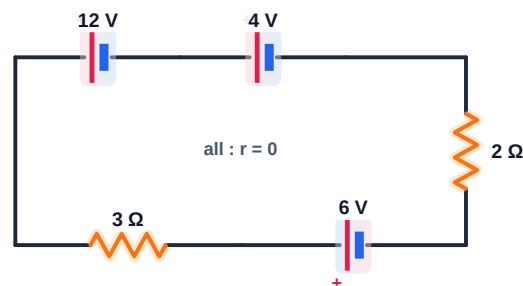


Fig. 59

- 92 A battery (12 V , 1Ω) and a battery (6 V , 1Ω) are connected in parallel (positive to positive) to a 4Ω resistor, as shown. The current in the 4Ω resistor is

MCQ

A 2 A

B 4 A

C 3.6 A

D 1 A

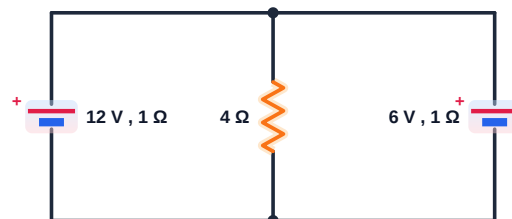


Fig. 60

- 93 In the circuit shown, the ammeter reads 2 A and the current flows in the direction shown. Each battery has an internal resistance of 1Ω . Find the emf V_{B2} .

Problem

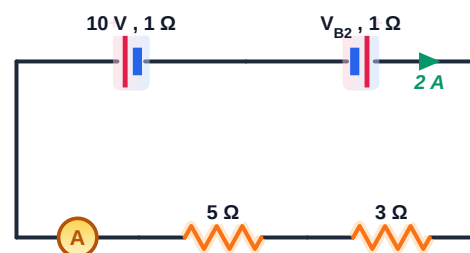


Fig. 61

94 For the circuit of Fig. 50, which equation is the correct loop equation for the left loop (the 20 V battery, the $2\ \Omega$ and the $4\ \Omega$ resistors)?

MCQ

A $20 + 2 I_1 - 4 I_3 = 0$

B $20 - 2 I_1 + 4 I_3 = 0$

C $20 - 2 I_1 - 4 I_3 = 0$

D $20 - 10 = 2 I_1 + 2 I_2$

95 In the circuit shown, A_1 reads 5 A and A_2 reads 2 A, while the lamp on the right carries 1.5 A. The reading of A_3 is

MCQ

A 3.5 A

B 1.5 A

C 3 A

D 0.5 A

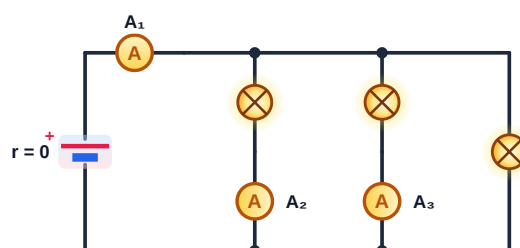


Fig. 62

96 In the circuit shown, the potential difference between point a and point b (the negative terminal of the battery) is

MCQ

A 8 V

B 4 V

C 12 V

D 2 V

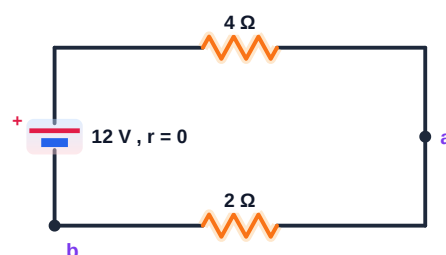


Fig. 63

97 In the bridge shown, the galvanometer reads zero. The resistance R equals

MCQ

A $60\ \Omega$

B $6.7\ \Omega$

C $15\ \Omega$

D $20\ \Omega$

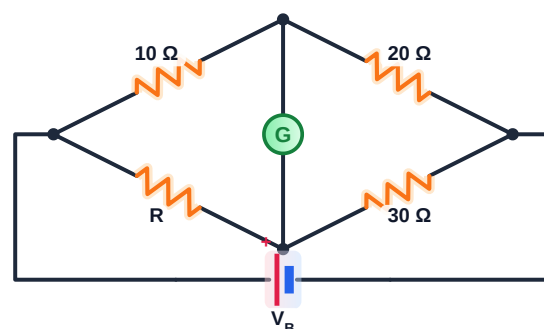


Fig. 64

98 In the circuit shown (both batteries have $r = 0$), the current in the middle branch xy is MCQ

A 3 A

B 1 A

C 2 A

D 4 A

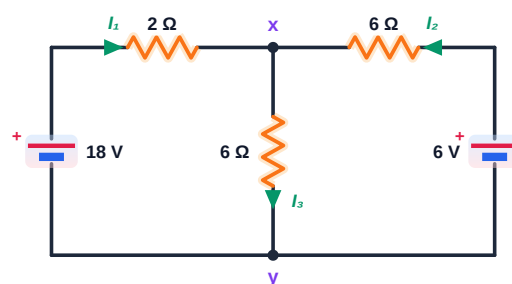


Fig. 65

99 A battery (12 V , 0.5 Ω) is used to charge a battery (9 V , 0.5 Ω) through a 2 Ω resistor. ★ HOTS Find: (i) the charging current, (ii) the terminal p.d. of the battery being charged, (iii) the power stored in it and the power turned into heat.

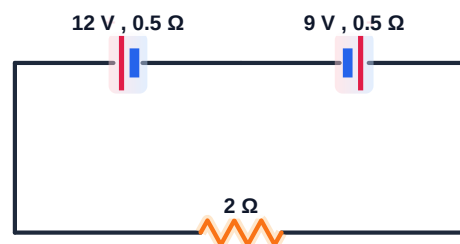


Fig. 66

100 In the circuit shown, the ammeter reads 0.5 A. The emf V_B of the battery is MCQ

A 9 V

B 6 V

C 3 V

D 4.5 V

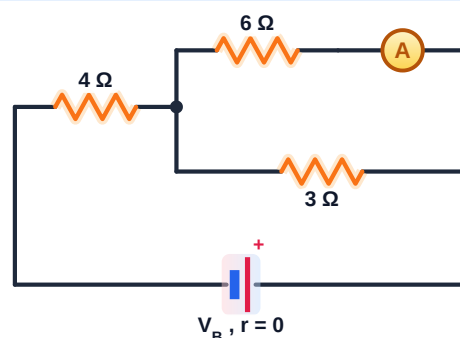


Fig. 67

Final Answers

Check yourself first — the full step-by-step solutions follow.

- 1 A) 4 A
- 2 A) 3
- 3 C) 1.875×10^{20}
- 4 B) 6 V
- 5 1.125×10^{21} electrons
- 6 C) 3
- 7 B) 2Ω
- 8 A) 0.68Ω
- 9 D) 3 A
- 10 B) R quadruples , σ unchanged
- 11 D) 80Ω
- 12 B) 27
- 13 (i) 44 m (ii) 110Ω (iii) 2 A
- 14 (i) $6 \times 10^{-8} \Omega \cdot m$ (ii) 15 m
- 15 A) R increases , I decreases , σ decreases
- 16 B) $c > d > a > b$
- 17 B) $4 \times 10^{-3} \Omega$
- 18 C) 0.5Ω
- 19 B) 25 W
- 20 A) 25 W
- 21 C) $1.32 \times 10^6 J$
- 22 C) $1/4$
- 23 A) a falling curve ($I \propto 1 / R_V$)
- 24 D) a horizontal line ($V = V_B$)
- 25 A) $4/3$
- 26 A) (1) , (3) , (4) , (2)
- 27 C) 10Ω
- 28 B) $2/3 A$
- 29 B) 4Ω
- 30 D) 3.6Ω
- 31 C) 8 V
- 32 A) 0.8 A
- 33 A) A increases , B decreases
- 34 B) the 60 W lamp
- 35 C) 4P
- 36 A) 6Ω
- 37 D) 13.3Ω
- 51 A) 8Ω
- 52 C) more than 12 V
- 53 B) 10 V
- 54 $V_B = 12 V$, $r = 1 \Omega$, $R = 5 \Omega$
- 55 $V_B = 12 V$, $r = 1 \Omega$
- 56 B) 3
- 57 $I = 3 A$; $V_1 = 4.5 V$, $V_2 = 6 V$
- 58 C) 7 V
- 59 A) 2Ω
- 60 C) 2 A
- 61 A) increases
- 62 C) 1 A
- 63 B) 1 A
- 64 D) 3Ω
- 65 C) zero
- 66 B) 5 A
- 67 considered : $r = 1 \Omega$
- 68 A) $1/2$
- 69 A) 540 J
- 70 $R_2 = 9 r$
- 71 A) 4 V , 2Ω
- 72 C) 7 V
- 73 C) 18 A
- 74 $R_V = 5 \Omega$
- 75 $r = 6 \Omega$
- 76 B) 5 A , from c to a
- 77 B) charge , energy
- 78 D) 3 A
- 79 B) $-I R$
- 80 $I_1 = 4 A$, $I_2 = -1 A$ (1 A the other way) , $I_3 = 3 A$
- 81 A) 36 W
- 82 B) it is being charged with a current of 1 A
- 83 C) 6 A
- 84 C) $-4 V$
- 85 B) $5R/6$
- 86 $I(CD) = 2 A$ from C to D ; $I = 10 A$; $R = 1.4 \Omega$
- 87 $I_1 = 4 A$, $I_2 = 2 A$, $R_2 = 12 \Omega$

38	D) 12 Ω	88	C) 8 V
39	B) 5/3	89	B) 1 A
40	I = 1.5 A ; I(6) = 1 A , I(12) = 0.5 A ; P = 9 W	90	B) 9/4
41	D) line 4	91	B) 2 A
42	A) from 1 A to 3 A	92	A) 2 A
43	R = 6 Ω	93	$V_{B2} = 30$ V
44	C) A decreases , B increases	94	C) $20 - 2 I_1 - 4 I_3 = 0$
45	A) 3 Ω	95	B) 1.5 A
46	B) 1.5 Ω	96	B) 4 V
47	n = 5 , R = 10 Ω	97	C) 15 Ω
48	A) ammeter reading increases , I(6 Ω) unchanged	98	C) 2 A
49	18 W , 12 W , 6 W ; total 36 W	99	I = 1 A ; V = 9.5 V ; 9 W stored , 3 W heat
50	A) lamp brighter , voltmeter reading increases	100	A) 9 V

Step-by-step solutions — Electric Current & Ohm's Law

Q 1 current from the number of electrons

$$Q = N e = 2 \times 10^{20} \times 1.6 \times 10^{-19} = 32 \text{ C}$$

$$I = Q / t = 32 / 8 = 4 \text{ A}$$

✓ A) 4 A

Q 2 slope of a Q–t graph

the slope of the Q – t graph is the current : $I = \Delta Q / \Delta t$

$$I(ab) = 12 / 2 = 6 \text{ A} \quad I(bc) = (18 - 12) / (5 - 2) = 2 \text{ A}$$

$$I(ab) / I(bc) = 6 / 2 = 3$$

✓ A) 3

Q 3 area under an I–t graph

the area under the I – t graph is the charge

$$Q = (4 \times 5) + (2 \times 5) = 20 + 10 = 30 \text{ C}$$

$$N = Q / e = 30 / 1.6 \times 10^{-19} = 1.875 \times 10^{20} \text{ electrons}$$

✓ C) 1.875×10^{20}

Q 4 potential difference

$$Q = N e = 5 \times 10^{19} \times 1.6 \times 10^{-19} = 8 \text{ C}$$

$$V = W / Q = 48 / 8 = 6 \text{ V}$$

✓ B) 6 V

Q 5 electrons through a filament

$$I = V / R = 12 / 4 = 3 \text{ A}$$

$$Q = I t = 3 \times 60 = 180 \text{ C}$$

$$N = Q / e = 180 / 1.6 \times 10^{-19} = 1.125 \times 10^{21} \text{ electrons}$$

✓ 1.125×10^{21} electrons

Q 6 slopes of V-I lines

on a V - I graph the slope is the resistance : $R = V / I = \tan \theta$
 line x makes 60° with the I-axis and line y makes 30°
 $R(x) / R(y) = \tan 60^\circ / \tan 30^\circ = 1.732 / 0.577 = 3$

✓ C) 3

Q 7 two graphs, one resistance

graph (1) : slope = $W / Q = V = 200 / 20 = 10 \text{ V}$
 graph (2) : slope = $Q / t = I = 20 / 4 = 5 \text{ A}$
 $R = V / I = 10 / 5 = 2 \Omega$

✓ B) 2 Ω **Q 8 resistance from resistivity**

$A = 0.5 \text{ mm}^2 = 0.5 \times 10^{-6} \text{ m}^2$
 $R = \rho_e L / A = 1.7 \times 10^{-8} \times 20 / 0.5 \times 10^{-6} = 0.68 \Omega$

✓ A) 0.68 Ω **Q 9 wire across a battery**

$A = \pi r^2 = 3.14 \times (10^{-3})^2 = 3.14 \times 10^{-6} \text{ m}^2$
 $R = \rho_e L / A = 5 \times 10^{-7} \times 6.28 / 3.14 \times 10^{-6} = 1 \Omega$
 $I = V / R = 3 / 1 = 3 \text{ A}$

✓ D) 3 A

Q 10 stretching a wire

constant volume : L doubles $\rightarrow$ A becomes $A / 2$
 $R = \rho_e L / A \rightarrow R' = \rho_e (2L) / (A / 2) = 4 R$
 the conductivity depends only on the material and the temperature $\rightarrow$ unchanged

✓ B) R quadruples , σ unchanged**Q 11 drawing a wire thinner**

$r \rightarrow r/2 \Rightarrow A \rightarrow A/4$, and constant volume $\Rightarrow L \rightarrow 4L$
 $R' / R = (L' / L) \times (A / A') = 4 \times 4 = 16$
 $R' = 16 \times 5 = 80 \Omega$

✓ D) 80 Ω **Q 12 resistance from length and mass**

$A = \text{volume} / L = (m / \rho) / L \Rightarrow R = \rho_e L / A = \rho_e \rho L^2 / m$
 $R \propto L^2 / m$ (same metal)
 $R(x) / R(y) = (9L^2 / 4) / (L^2 / 12) = (9 / 4) \times 12 = 27$

✓ B) 27

Q 13 a wire wound into a coil

(i) $L = 2 \pi r N = 2 \times (22 / 7) \times 0.07 \times 100 = 44 \text{ m}$
 (ii) $R = \rho_e L / A = 5 \times 10^{-7} \times 44 / 0.2 \times 10^{-6} = 110 \Omega$
 (iii) $I = V / R = 220 / 110 = 2 \text{ A}$

✓ (i) 44 m (ii) 110 Ω (iii) 2 A

Q 14 resistivity from an R-L graph

(i) slope = $R / L = 12 / 40 = 0.3 \, \Omega/\text{m}$ and $R = (\rho_e / A) L \Rightarrow \text{slope} = \rho_e / A$

$$\rho_e = \text{slope} \times A = 0.3 \times 0.2 \times 10^{-6} = 6 \times 10^{-8} \, \Omega \cdot \text{m}$$

(ii) $L = R / \text{slope} = 4.5 / 0.3 = 15 \, \text{m}$

✓ (i) $6 \times 10^{-8} \, \Omega \cdot \text{m}$ (ii) 15 m

Q 15 effect of temperature

a higher temperature makes the atoms vibrate more $\rightarrow$ more collisions with the free electrons
so the resistivity rises $\rightarrow R$ increases and $\sigma = 1 / \rho_e$ decreases

V is constant $\rightarrow I = V / R$ decreases

✓ A) R increases , I decreases , σ decreases

Q 16 ranking four wires

$$I = V / R = V A / (\rho_e L) \Rightarrow I \propto A / L$$

$$a : A / L = 1 \quad b : A / 2L = 0.5 \quad c : 2A / L = 2 \quad d : 3A / 2L = 1.5$$

$$c > d > a > b$$

✓ B) $c > d > a > b$

Q 17 a hollow tube

the current flows through the ring-shaped area only

$$A = \pi (r_2^2 - r_1^2) = 3.14 \times (9 - 4) \times 10^{-6} = 1.57 \times 10^{-5} \, \text{m}^2$$

$$R = \rho_e L / A = 3.14 \times 10^{-8} \times 2 / 1.57 \times 10^{-5} = 4 \times 10^{-3} \, \Omega$$

✓ B) $4 \times 10^{-3} \, \Omega$

Q 18 a semicircular cross-section

$$r = 2 \, \text{mm} , \text{ the area is half a circle : } A = \pi r^2 / 2 = 3.14 \times 4 \times 10^{-6} / 2 = 6.28 \times 10^{-6} \, \text{m}^2$$

$$R = \rho_e L / A = 2 \times 10^{-6} \times 1.57 / 6.28 \times 10^{-6} = 0.5 \, \Omega$$

✓ C) 0.5 Ω

Q 19 a lamp on half its rated voltage

$$R = V^2 / P = 220^2 / 100 = 484 \, \Omega$$

$$P' = V'^2 / R = 110^2 / 484 = 25 \, \text{W} \text{ (half the voltage } \rightarrow \text{ a quarter of the power)}$$

✓ B) 25 W

Q 20 slope of a P-I² graph

$$P = I^2 R \Rightarrow \text{the slope of the } P - I^2 \text{ graph is } R = 36 / 9 = 4 \, \Omega$$

$$P = V^2 / R = 10^2 / 4 = 25 \, \text{W}$$

✓ A) 25 W

Q 21 energy of a heater

$$P = V^2 / R = 220^2 / 22 = 2200 \, \text{W}$$

$$W = P t = 2200 \times (10 \times 60) = 1.32 \times 10^6 \, \text{J}$$

✓ C) $1.32 \times 10^6 \, \text{J}$

Q 22 power in parallel wires

same V : $P = V^2 / R \Rightarrow P \propto 1 / R$

$R \propto 1 / r^2 \Rightarrow P \propto r^2$

$P_1 / P_2 = (1 / 2)^2 = 1 / 4$

✓ C) 1/4

Q 23 ammeter reading and R_V

$r = 0 \Rightarrow$ the p.d. across the rheostat always equals V_B

$I = V_B / R_V \Rightarrow I$ is inversely proportional to R_V (a falling curve)

✓ A) a falling curve ($I \propto 1 / R_V$)

Q 24 voltmeter reading and R_V

the voltmeter is connected across the rheostat, and the ammeter has negligible resistance

$r = 0 \Rightarrow V = V_B - I r = V_B$ whatever R_V is $\rightarrow$ a horizontal line

✓ D) a horizontal line ($V = V_B$)

Q 25 conductivity, length and area together

$R = L / (\sigma A) \Rightarrow I = V / R = V \sigma A / L \Rightarrow I \propto \sigma A / L$

$x : \sigma A / L \quad y : (\sigma / 2) (3A) / (2L) = 3 \sigma A / 4L$

$I(x) / I(y) = 1 / (3 / 4) = 4 / 3$

✓ A) 4/3

Step-by-step solutions — Connecting Resistors

Q 26 four ways of connecting four resistors

(1) all in series : $4R$

(3) two parallel pairs in series : $R/2 + R/2 = R$

(4) $3R$ in parallel with R : $(3R \times R) / 4R = 3R/4$

(2) all in parallel : $R/4$

$4R > R > 3R/4 > R/4$

✓ A) (1), (3), (4), (2)

Q 27 a wire cut into six parts

each part : $90 / 6 = 15 \, \Omega$

three equal parts in parallel : $15 / 3 = 5 \, \Omega$

the two groups are in series : $5 + 5 = 10 \, \Omega$

✓ C) $10 \, \Omega$

Q 28 current divider

$R' = (6 \times 3) / (6 + 3) = 2 \, \Omega \Rightarrow V = I R' = 2 \times 2 = 4 \, V$

$I_1 = V / 6 = 4 / 6 = 2/3 \, A$ (and $I_2 = 4 / 3 \, A$ through the $3 \, \Omega$)

✓ B) $2/3 \, A$

Q 29 a short-circuited resistor

the wire joins the two ends of $4\ \Omega$: no p.d. across it $\rightarrow$ it carries no current and is ignored
 $6\ \Omega // 3\ \Omega = (6 \times 3) / 9 = 2\ \Omega$
 $R(ab) = 2 + 2 = 4\ \Omega$

✓ B) $4\ \Omega$ **Q 30 a balanced bridge**

$2 / 4 = 3 / 6 = 0.5 \Rightarrow$ the two middle points have the same potential
 so the $10\ \Omega$ resistor carries no current and is ignored
 $R(ab) = (2 + 4) // (3 + 6) = (6 \times 9) / 15 = 3.6\ \Omega$

✓ D) $3.6\ \Omega$ **Q 31 voltmeter in a series circuit**

series : $R = 2 + 4 = 6\ \Omega \Rightarrow I = 12 / 6 = 2\ \text{A}$
 $V(4\ \Omega) = I R = 2 \times 4 = 8\ \text{V}$

✓ C) $8\ \text{V}$ **Q 32 from a voltmeter reading to a branch current**

the voltmeter is across the $2\ \Omega$ resistor : $I = 4 / 2 = 2\ \text{A}$ (the total current)
 this current divides between $6\ \Omega$ and $4\ \Omega$:
 $I(6) = I \times 4 / (6 + 4) = 2 \times 0.4 = 0.8\ \text{A}$

✓ A) $0.8\ \text{A}$ **Q 33 closing a switch in a lamp network**

K open : $B // C = R/2$, total = $1.5 R$, $I = V / 1.5R = 0.67\ \text{V}/R$
 K closed : $B // C // D = R/3$, total = $4R/3$, $I = 0.75\ \text{V}/R \rightarrow A$ (carries I) is brighter
 p.d. across the group : before $V/3$, after $I \times R/3 = V/4 \rightarrow B$ is dimmer

✓ A) A increases , B decreases

Q 34 rated lamps in series

$R = V^2 / P$: $R(100) = 484\ \Omega$ $R(60) = 807\ \Omega$
 in series both carry the same current $\Rightarrow P = I^2 R \propto R$
 the $60\ \text{W}$ lamp (larger R) consumes more power $\rightarrow$ it is brighter

✓ B) the $60\ \text{W}$ lamp**Q 35 series vs parallel power**

series : $2R \Rightarrow P = V^2 / 2R$
 parallel : $R/2 \Rightarrow P' = V^2 / (R/2) = 2V^2 / R$
 $P' / P = (2 / R) / (1 / 2R) = 4 \Rightarrow P' = 4P$

✓ C) $4P$ **Q 36 a mixed network**

$4\ \Omega$ and $8\ \Omega$ are in series : $12\ \Omega$
 $12\ \Omega // 6\ \Omega = (12 \times 6) / 18 = 4\ \Omega$
 $R(ab) = 2 + 4 = 6\ \Omega$

✓ A) $6\ \Omega$

Q 37 two conductors in contact

$$R = L / (\sigma A) : R(x) = L / (4 \sigma A) = 5 \, \Omega \Rightarrow L / (\sigma A) = 20 \, \Omega$$

$$\text{each half of } y : L / (\sigma \times 2A) = 10 \, \Omega$$

$$\text{the first half of } y \text{ is in parallel with } x : (5 \times 10) / 15 = 3.33 \, \Omega$$

$$\text{then the second half of } y \text{ in series : } R(ab) = 3.33 + 10 = 13.3 \, \Omega$$

✓ D) 13.3 Ω **Q 38 possible combinations**

all in series : 18 Ω all in parallel : 2 Ω

$$6 + (6 // 6) = 6 + 3 = 9 \, \Omega$$

$$6 // (6 + 6) = (6 \times 12) / 18 = 4 \, \Omega$$

12 Ω needs only two resistors in series $\rightarrow$ it cannot be made with all three

✓ D) 12 Ω **Q 39 two voltmeters on one series line**

series : the same current I flows through R , $2R$ and $3R$

$$V_1 \text{ is across } 2R + 3R = 5R \Rightarrow V_1 = 5 I R$$

$$V_2 \text{ is across } R + 2R = 3R \Rightarrow V_2 = 3 I R$$

$$V_1 / V_2 = 5 / 3$$

✓ B) 5/3

Q 40 solving a whole circuit

$$(i) 6 // 12 = (6 \times 12) / 18 = 4 \, \Omega \Rightarrow R = 4 + 4 = 8 \, \Omega \Rightarrow I = 12 / 8 = 1.5 \, A$$

$$(ii) V(\text{parallel}) = 1.5 \times 4 = 6 \, V \Rightarrow I(6) = 6 / 6 = 1 \, A, I(12) = 6 / 12 = 0.5 \, A$$

$$(iii) P = I^2 R = 1.5^2 \times 4 = 9 \, W$$

✓ I = 1.5 A ; I(6) = 1 A , I(12) = 0.5 A ; P = 9 W

Q 41 a parallel combination on a V-I graph

$$R = \text{slope} : x = 9 / 1 = 9 \, \Omega \quad y = 9 / 3 = 3 \, \Omega$$

parallel : $(9 \times 3) / 12 = 2.25 \, \Omega$ (smaller than both, so below both lines)

line 4 passes through $(2 \, A, 4.5 \, V)$: slope = $4.5 / 2 = 2.25 \, \Omega \rightarrow$ line 4

✓ D) line 4

Q 42 a switch across a resistor

$$K \text{ open} : R = 4 + 8 = 12 \, \Omega \Rightarrow I = 12 / 12 = 1 \, A$$

$$K \text{ closed} : \text{the switch short-circuits the } 8 \, \Omega \text{ resistor} \Rightarrow R = 4 \, \Omega$$

$$I = 12 / 4 = 3 \, A$$

✓ A) from 1 A to 3 A

Q 43 finding an unknown resistance

$$\text{total resistance} = V / I = 24 / 4 = 6 \, \Omega$$

$$\text{the parallel group} = 6 - 2 = 4 \, \Omega$$

$$1 / R + 1 / 12 = 1 / 4 \Rightarrow 1 / R = 3/12 - 1/12 = 2/12$$

$$R = 6 \, \Omega$$

✓ R = 6 Ω

Q 44 a lamp burns out

before : $B \parallel C = R/2$, total $1.5R$, $I = V / 1.5R \rightarrow B$ carries $V / 3R$

after : A and B in series , total $2R$, $I = V / 2R$

A : $V/1.5R \rightarrow V/2R$ (less current, dimmer) B : $V/3R \rightarrow V/2R$ (more current, brighter)

✓ C) A decreases , B increases

Q 45 a ring of wire

the short arc is $1/4$ of the wire : 4Ω the long arc is $3/4$: 12Ω

the two arcs are in parallel between a and b

$$R(ab) = (4 \times 12) / 16 = 3 \Omega$$

✓ A) 3Ω

Q 46 redraw before you calculate

follow the wires : every resistor has one end joined to a and the other end joined to b
so the three resistors are all in parallel

$$1/R = 1/6 + 1/3 + 1/6 = 4/6 \Rightarrow R = 1.5 \Omega$$

✓ B) 1.5Ω

Q 47 series and parallel together

series : $n R = 50$ parallel : $R / n = 2$

multiply : $R^2 = 100 \Rightarrow R = 10 \Omega$

$$n = 50 / 10 = 5$$

✓ $n = 5$, $R = 10 \Omega$

Q 48 adding a parallel branch

$r = 0$: each branch has the full 12 V across it

$I(6) = 12 / 6 = 2 \text{ A}$ before and after closing K (unchanged)

K closed : the 3Ω branch adds $12 / 3 = 4 \text{ A} \rightarrow$ ammeter : $2 \text{ A} \rightarrow 6 \text{ A}$

✓ A) ammeter reading increases , $I(6 \Omega)$ unchanged

Q 49 power in parallel

in parallel each resistor has $V = 6 \text{ V}$: $P = V^2 / R$

$$P(2) = 36 / 2 = 18 \text{ W} \quad P(3) = 36 / 3 = 12 \text{ W} \quad P(6) = 36 / 6 = 6 \text{ W}$$

total = $18 + 12 + 6 = 36 \text{ W}$ (check : $R' = 1 \Omega$, $P = 36 / 1 = 36 \text{ W}$)

✓ 18 W , 12 W , 6 W ; total 36 W

Q 50 a rheostat in series with a lamp

R_v decreases $\rightarrow$ the total resistance decreases $\rightarrow$ the current increases

the lamp carries a larger current $\rightarrow$ brighter

the voltmeter is across the lamp : $V = I R(\text{lamp})$ increases

✓ A) lamp brighter , voltmeter reading increases

Q 51 a triangle of wire

each side : $36 / 3 = 12 \Omega$

between x and y : one side (12Ω) in parallel with two sides (24Ω)

$$R = (12 \times 24) / 36 = 8 \Omega$$

✓ A) 8Ω

Step-by-step solutions — Ohm's Law for the Closed Circuit

Q 52 charging a battery

while charging, the current enters the battery through its positive terminal

$$V = V_B + I r \Rightarrow V > 12 \text{ V}$$

$$(\text{while discharging } V = V_B - I r < V_B)$$

✓ C) more than 12 V

Q 53 terminal potential difference

$$I = V_B / (R + r) = 12 / (5 + 1) = 2 \text{ A}$$

$$\text{the voltmeter is across the battery terminals : } V = V_B - I r = 12 - 2 \times 1 = 10 \text{ V}$$

$$(= I R = 2 \times 5 = 10 \text{ V})$$

✓ B) 10 V

Q 54 open and closed switch

$$(i) \text{ K open : no current, so the voltmeter reads the emf : } V_B = 12 \text{ V}$$

$$(ii) \text{ K closed : } V = V_B - I r \Rightarrow 10 = 12 - 2 r \Rightarrow r = 1 \Omega$$

$$(iii) R = V / I = 10 / 2 = 5 \Omega$$

✓ $V_B = 12 \text{ V}, r = 1 \Omega, R = 5 \Omega$

Q 55 two resistors, one battery

$$V_B = I_1 (R_1 + r) = 2 (5 + r)$$

$$V_B = I_2 (R_2 + r) = 1 (11 + r)$$

$$10 + 2 r = 11 + r \Rightarrow r = 1 \Omega$$

$$V_B = 2 (5 + 1) = 12 \text{ V}$$

✓ $V_B = 12 \text{ V}, r = 1 \Omega$

Q 56 terminal p.d. as a fraction of the emf

$$\text{with } R \text{ connected : } V = I R = V_0 R / (R + r)$$

$$R / (R + r) = 0.75 \Rightarrow R = 0.75 R + 0.75 r \Rightarrow 0.25 R = 0.75 r$$

$$R / r = 3$$

✓ B) 3

Q 57 batteries in series

$$(i) I = (6 + 9) / (3.5 + 0.5 + 1) = 15 / 5 = 3 \text{ A}$$

$$(ii) \text{ both discharge : } V_1 = 6 - 3 \times 0.5 = 4.5 \text{ V } V_2 = 9 - 3 \times 1 = 6 \text{ V}$$

$$\text{check : } 4.5 + 6 = 10.5 \text{ V} = 3 \times 3.5 \checkmark$$

✓ $I = 3 \text{ A}; V_1 = 4.5 \text{ V}, V_2 = 6 \text{ V}$

Q 58 two batteries in opposition

$$\text{opposing : } I = (12 - 6) / (4 + 1 + 1) = 6 / 6 = 1 \text{ A}$$

the current enters the 6 V battery through its positive terminal → it is being charged

$$V = V_B + I r = 6 + 1 \times 1 = 7 \text{ V}$$

✓ C) 7 V

Q 59 a terminal-p.d. graph

$V = V_B - I r$: the intercept on the V-axis is $V_B = 12 \text{ V}$
 the slope is $-r$: $r = 12 / 6 = 2 \Omega$
 (6 A is the short-circuit current V_B / r)

✓ A) 2Ω **Q 60 using the graph**

from the graph : $V_B = 12 \text{ V}$, $r = 2 \Omega$
 $I = V_B / (R + r) = 12 / (4 + 2) = 2 \text{ A}$

✓ C) 2 A **Q 61 the rheostat and the terminal p.d.**

$R_v \uparrow \rightarrow I = V_B / (R_v + r)$ decreases
 $V = V_B - I r$: a smaller $I r$ is lost inside the battery
 so V increases (towards V_B)

✓ A) increases

Q 62 cells in series

total emf = $4 \times 1.5 = 6 \text{ V}$ total internal resistance = $4 \times 0.5 = 2 \Omega$
 $I = 6 / (4 + 2) = 1 \text{ A}$

✓ C) 1 A **Q 63 a reversed cell**

the reversed cell cancels one correct cell : emf = $(5 - 1) \times 2 = 8 \text{ V}$
 the internal resistances all add up : $6 \times 0.5 = 3 \Omega$
 $I = 8 / (5 + 3) = 1 \text{ A}$

✓ B) 1 A **Q 64 a battery feeding a bridge**

K open : $V_B = 12 \text{ V}$ K closed : $9 = 12 - I \times 1 \Rightarrow I = 3 \text{ A}$
 external resistance = $9 / 3 = 3 \Omega$
 the bridge is balanced ($R / R = R / R$) $\rightarrow$ the middle resistor carries no current
 network = $(R + R) // (R + R) = R \Rightarrow R = 3 \Omega$

✓ D) 3Ω **Q 65 a short circuit**

short circuit : $R = 0 \Rightarrow I = V_B / r$
 $V = V_B - I r = V_B - (V_B / r) r = 0$

✓ C) zero

Q 66 an ammeter that reads zero

A_2 reads zero $\Rightarrow$ balanced : $2 / 4 = 3 / R \Rightarrow R = 6 \Omega$
 branches : $2 + 4 = 6 \Omega$ and $3 + 6 = 9 \Omega \Rightarrow (6 \times 9) / 15 = 3.6 \Omega$
 $I = V_B / (R' + r) = 23 / (3.6 + 1) = 5 \text{ A}$

✓ B) 5 A

Q 67 is r neglected?

$$V(6\ \Omega) = 1 \times 6 = 6\text{ V} = V(3\ \Omega) \Rightarrow I(3) = 6 / 3 = 2\text{ A}$$

$$\text{total current } I = 1 + 2 = 3\text{ A} \Rightarrow V(2\ \Omega) = 3 \times 2 = 6\text{ V}$$

external p.d. = $6 + 6 = 12\text{ V} < 15\text{ V} \rightarrow 3\text{ V}$ is lost inside the battery

$$r = 3 / 3 = 1\ \Omega \rightarrow \text{the internal resistance is considered}$$

✓ considered : $r = 1\ \Omega$

Q 68 discharging vs charging

(1) the current leaves through the positive terminal $\rightarrow$ discharging : $V = 12 - 2 \times 2 = 8\text{ V}$

(2) the current enters through the positive terminal $\rightarrow$ charging : $V = 12 + 2 \times 2 = 16\text{ V}$

$$V(xy) / V(zk) = 8 / 16 = 1/2$$

✓ A) $1/2$

Q 69 energy lost inside a battery

$$I = 24 / (7 + 1) = 3\text{ A}$$

$$W = I^2 r t = 3^2 \times 1 \times 60 = 540\text{ J}$$

✓ A) 540 J

Q 70 two voltmeters

the same current I flows through R_1 and R_2

$$V(\text{terminal}) = I (R_1 + R_2) \quad V_1 = I R_1$$

$$V = 4 V_1 \Rightarrow R_1 + R_2 = 4 R_1 \Rightarrow R_2 = 3 R_1$$

$$R_2 = 3 \times 3r = 9r$$

✓ $R_2 = 9r$

Q 71 a $1/I - R$ graph

$$V_B = I (R + r) \Rightarrow 1 / I = R / V_B + r / V_B$$

$$\text{slope} = 1 / V_B = (3 - 0.5) / 10 = 0.25 \Rightarrow V_B = 4\text{ V}$$

$$\text{intercept} = r / V_B = 0.5 \Rightarrow r = 0.5 \times 4 = 2\ \Omega$$

✓ A) 4 V, 2 Ω

Q 72 the charging voltage

$$\text{charging : } V = V_B + I r = 6 + 2 \times 0.5 = 7\text{ V}$$

✓ C) 7 V

Q 73 the short-circuit current

short circuit : $R = 0$

$$I(\text{max}) = V_B / r = 9 / 0.5 = 18\text{ A}$$

✓ C) 18 A

Q 74 making a lamp work normally

$$\text{normal working : } I = P / V = 3 / 6 = 0.5\text{ A}, R(\text{lamp}) = V^2 / P = 36 / 3 = 12\ \Omega$$

$$V_B = I (R(\text{lamp}) + R_v + r) \Rightarrow 9 = 0.5 (12 + R_v + 1)$$

$$18 = 13 + R_v \Rightarrow R_v = 5\ \Omega$$

✓ $R_v = 5\ \Omega$

Q 75 equal power in two resistors

$$P = I^2 R = V_B^2 R / (R + r)^2$$

$$4 / (4 + r)^2 = 9 / (9 + r)^2 \Rightarrow 2 / (4 + r) = 3 / (9 + r)$$

$$18 + 2r = 12 + 3r \Rightarrow r = 6 \Omega$$

$$\text{check : } 4 / 100 = 9 / 225 = 0.04 \checkmark$$

✓ **r = 6 Ω****Step-by-step solutions — Kirchhoff's Laws****Q 76** the junction ruleKirchhoff's first law : $\Sigma I(\text{in}) = \Sigma I(\text{out})$ known : $\text{in} = 5 + 4 = 9 \text{ A}$ $\text{out} = 8 + 6 = 14 \text{ A}$ the missing 5 A must flow **into** a : $I = 5 \text{ A}$, from c to a✓ **B) 5 A , from c to a****Q 77** the principles behind the two laws

1st law : charge does not pile up at a junction → conservation of charge

2nd law : the energy given by the sources in a loop = the energy used → conservation of energy

✓ **B) charge , energy****Q 78** junction and parallel branches $V = 2 \times 6 = 12 \text{ V}$ across both resistors (parallel)

$$I(12 \Omega) = 12 / 12 = 1 \text{ A}$$

$$\text{junction : } I = 2 + 1 = 3 \text{ A}$$

✓ **D) 3 A****Q 79** sign convention

moving with the current through a resistor, the potential falls (from high to low)

so the term is $- I R$ (and $+ I R$ if we move against the current)✓ **B) $- I R$** **Q 80** a two-loop circuit

$$\text{junction x : } I_1 + I_2 = I_3$$

$$\text{left loop : } 20 - 2 I_1 - 4 I_3 = 0$$

$$\text{right loop : } 10 - 2 I_2 - 4 I_3 = 0$$

$$\text{subtract the loop equations : } 10 - 2 I_1 + 2 I_2 = 0 \Rightarrow I_1 = I_2 + 5$$

$$\text{left loop : } 20 - 2 (I_2 + 5) - 4 (2 I_2 + 5) = 0 \Rightarrow -10 I_2 - 10 = 0$$

$$I_2 = -1 \text{ A} , I_1 = 4 \text{ A} , I_3 = 3 \text{ A}$$

✓ **$I_1 = 4 \text{ A}$, $I_2 = -1 \text{ A}$ (1 A the other way) , $I_3 = 3 \text{ A}$** **Q 81** power in the middle branchfrom question 80 : $I_3 = 3 \text{ A}$

$$P = I^2 R = 3^2 \times 4 = 36 \text{ W}$$

✓ **A) 36 W**

Q 82 which battery is charged?

$I_2 = -1 \text{ A}$: the real current is opposite to the arrow drawn
 so 1 A enters the 10 V battery through its positive terminal → it is being charged

✓ B) it is being charged with a current of 1 A

Q 83 two batteries feeding one resistor

let the potential of x (with y = 0) be V :

$$(24 - V) / 4 + (12 - V) / 4 = V / 1 \Rightarrow 36 - 2V = 4V \Rightarrow V = 6 \text{ V}$$

$$I(1 \Omega) = 6 / 1 = 6 \text{ A} \quad (I_1 = 4.5 \text{ A}, I_2 = 1.5 \text{ A} : 4.5 + 1.5 = 6 \checkmark)$$

✓ C) 6 A

Q 84 walking along a branch

walk from a to b : through 2Ω with the current → $-3 \times 2 = -6 \text{ V}$

through the battery from - to + → $+10 \text{ V}$

$$V_b = V_a - 6 + 10 \Rightarrow V_a - V_b = -4 \text{ V}$$

✓ C) -4 V

Q 85 a cube of resistors

by symmetry the current I entering a divides equally into 3 edges : $I/3$ each

at the next corners each current divides into 2 : $I/6$ on each of the 6 middle edges

the last 3 edges carry $I/3$ each into b

$$V(ab) = (I/3)R + (I/6)R + (I/3)R = 5IR/6 \Rightarrow R(ab) = 5R/6$$

✓ B) $5R/6$

Q 86 an unbalanced bridge

take $V(B) = 0$, $V(A) = 14 \text{ V}$, and the unknown potentials $V(C)$, $V(D)$

$$\text{junction C : } (14 - C) / 1 = (C - D) / 1 + C / 2$$

$$\text{junction D : } (14 - D) / 2 + (C - D) / 1 = D / 1$$

$$\text{solving : } V(C) = 8 \text{ V}, V(D) = 6 \text{ V}$$

$$(i) I(CD) = (8 - 6) / 1 = 2 \text{ A, from C to D}$$

$$(ii) I = I(AC) + I(AD) = (14 - 8) / 1 + (14 - 6) / 2 = 6 + 4 = 10 \text{ A}$$

$$(iii) R = 14 / 10 = 1.4 \Omega$$

✓ I(CD) = 2 A from C to D ; I = 10 A ; R = 1.4 Ω

Q 87 from one known current

$$\text{loop (battery, } 2 \Omega, 6 \Omega) : 36 - 6 \times 2 - 6 I_1 = 0 \Rightarrow I_1 = 4 \text{ A}$$

$$\text{junction : } 6 = I_1 + I_2 \Rightarrow I_2 = 2 \text{ A}$$

$$\text{the } 6 \Omega \text{ and } R_2 \text{ have the same p.d. : } 24 \text{ V} \Rightarrow R_2 = 24 / 2 = 12 \Omega$$

✓ $I_1 = 4 \text{ A}, I_2 = 2 \text{ A}, R_2 = 12 \Omega$

Q 88 a galvanometer that reads zero

G reads zero → no current in the lower loop, so V_{B2} equals the p.d. across 8Ω

$$\text{upper loop : } I = 12 / (4 + 8) = 1 \text{ A}$$

$$V(8 \Omega) = 1 \times 8 = 8 \text{ V} \Rightarrow V_{B2} = 8 \text{ V}$$

✓ C) 8 V

Q 89 identical batteries in parallel

identical batteries in parallel $\rightarrow$ they share the current equally : $I_1 = I_2 = I / 2$
 loop (one battery + the resistor) : $6 - 1 \times (I / 2) - 2.5 I = 0 \Rightarrow 3 I = 6$
 $I = 2 \text{ A}$ in the resistor $\Rightarrow 1 \text{ A}$ in each battery

✓ B) 1 A

Q 90 adding sources in parallel

let V be the p.d. across S . Each connected branch gives $(V_B - V) / R$ and S takes V / R
 both open (1 branch) : $V_B - V = V \Rightarrow V = V_B / 2$
 both closed (3 branches) : $3 (V_B - V) = V \Rightarrow V = 3 V_B / 4$
 $P \propto V^2$: $n = (3/4)^2 / (1/2)^2 = 9/4$

✓ B) 9/4

Q 91 one loop, three batteries

the 12 V and 4 V batteries push the current the same way, the 6 V battery opposes them
 $\Sigma V_B = \Sigma I R$: $12 + 4 - 6 = I (2 + 3)$
 $I = 10 / 5 = 2 \text{ A}$

✓ B) 2 A

Q 92 unequal batteries in parallel

let V be the p.d. across the 4Ω resistor :
 $(12 - V) / 1 + (6 - V) / 1 = V / 4 \Rightarrow 72 - 8V = V \Rightarrow V = 8 \text{ V}$
 $I(4 \Omega) = 8 / 4 = 2 \text{ A}$ (the 12 V battery gives 4 A , the 6 V battery is charged by 2 A)

✓ A) 2 A

Q 93 finding an unknown emf

the two batteries oppose each other and the current flows the way V_{B2} pushes it
 loop : $V_{B2} - 10 = I (3 + 5 + 1 + 1) = 2 \times 10 = 20$
 $V_{B2} = 30 \text{ V}$ (the 10 V battery is being charged)

✓ $V_{B2} = 30 \text{ V}$ **Q 94 writing a loop equation**

go round the left loop in the direction of I_1 :
 through the battery from $-$ to $+$: $+ 20$; through 2Ω with I_1 : $- 2 I_1$; down through 4Ω with I_3 : $- 4 I_3$
 $20 - 2 I_1 - 4 I_3 = 0$ (check : $20 - 8 - 12 = 0 \checkmark$)

✓ C) $20 - 2 I_1 - 4 I_3 = 0$ **Q 95 the junction rule with ammeters**

the main current 5 A divides into the three lamp branches
 $5 = 2 + I(A_3) + 1.5 \Rightarrow I(A_3) = 1.5 \text{ A}$

✓ B) 1.5 A

Q 96 potential of a point

$$I = 12 / (4 + 2) = 2 \text{ A}$$

from a to b the current passes only through the 2Ω resistor

$$V(ab) = I \times 2 = 4 \text{ V}$$

✓ B) 4 V

Q 97 a balanced bridge with a galvanometer

G reads zero → the two ends of G are at the same potential (balanced bridge)

$$10 / 20 = R / 30$$

$$R = 15 \Omega$$

✓ C) 15 Ω

Q 98 another two-loop circuit

let V be the potential of x ($y = 0$) :

$$(18 - V) / 2 + (6 - V) / 6 = V / 6 \Rightarrow 54 - 3V + 6 - V = V \Rightarrow V = 12 \text{ V}$$

$$I(\text{middle}) = 12 / 6 = 2 \text{ A} \quad (I_1 = 3 \text{ A} ; \text{ the } 6 \text{ V battery is charged by } 1 \text{ A})$$

✓ C) 2 A

Q 99 one battery charging another

$$(i) I = (12 - 9) / (2 + 0.5 + 0.5) = 3 / 3 = 1 \text{ A}$$

$$(ii) \text{ the } 9 \text{ V battery is charged : } V = 9 + 1 \times 0.5 = 9.5 \text{ V}$$

$$(iii) \text{ stored : } V_B I = 9 \times 1 = 9 \text{ W heat : } I^2 (R + r_1 + r_2) = 1 \times 3 = 3 \text{ W}$$

$$(\text{ the } 12 \text{ V battery gives } 12 \times 1 = 12 \text{ W} = 9 + 3 \checkmark)$$

✓ I = 1 A ; V = 9.5 V ; 9 W stored , 3 W heat

Q 100 working backwards to the emf

$$V(6 \Omega) = 0.5 \times 6 = 3 \text{ V} = V(3 \Omega) \Rightarrow I(3) = 3 / 3 = 1 \text{ A}$$

$$\text{junction : } I = 0.5 + 1 = 1.5 \text{ A through the } 4 \Omega$$

$$\text{loop : } V_B = 1.5 \times 4 + 3 = 9 \text{ V}$$

✓ A) 9 V